

Personal Loan Campaign - AllLife Bank

Personal Loan Campaign for AI & ML

Contents/Agenda

- Executive Summary
- Business Problem Overview and Solution Approach
- EDA Results
- Data Preprocessing
- Model Performance Summary

Executive Summary - Insights

- Income is the most dominant predictor in the data
- CD Account holders are low hanging fruit and have a high conversion rate to personal loans
- CCAvg and Family size are other important features
 - Higher monthly credit card spend and larger family sizes are great predictors for personal loan candidates
- Factors like Age and Professional experience show barely any statistical difference between borrows and non borrowers and should not be a focus for the company when identifying potential borrowers

Executive Summary - Recommendations

- The bank should focus on depositors with an annual income of over \$100,000 and that have a family size over 2.5. These are great prospects
- Borrowers with a high Mortgage may need personal loans to improve their homes and are great candidates to target
- Use the post-pruned model to target the groups above and this will reduce the time used on half of the depositor base and keep potential revenue at around 99%
- This then reduces the budget that was used to go after potential depositors that are now identified as weak leads that will not convert. Now you can focus the budget on strong leads only now that these leads are identified

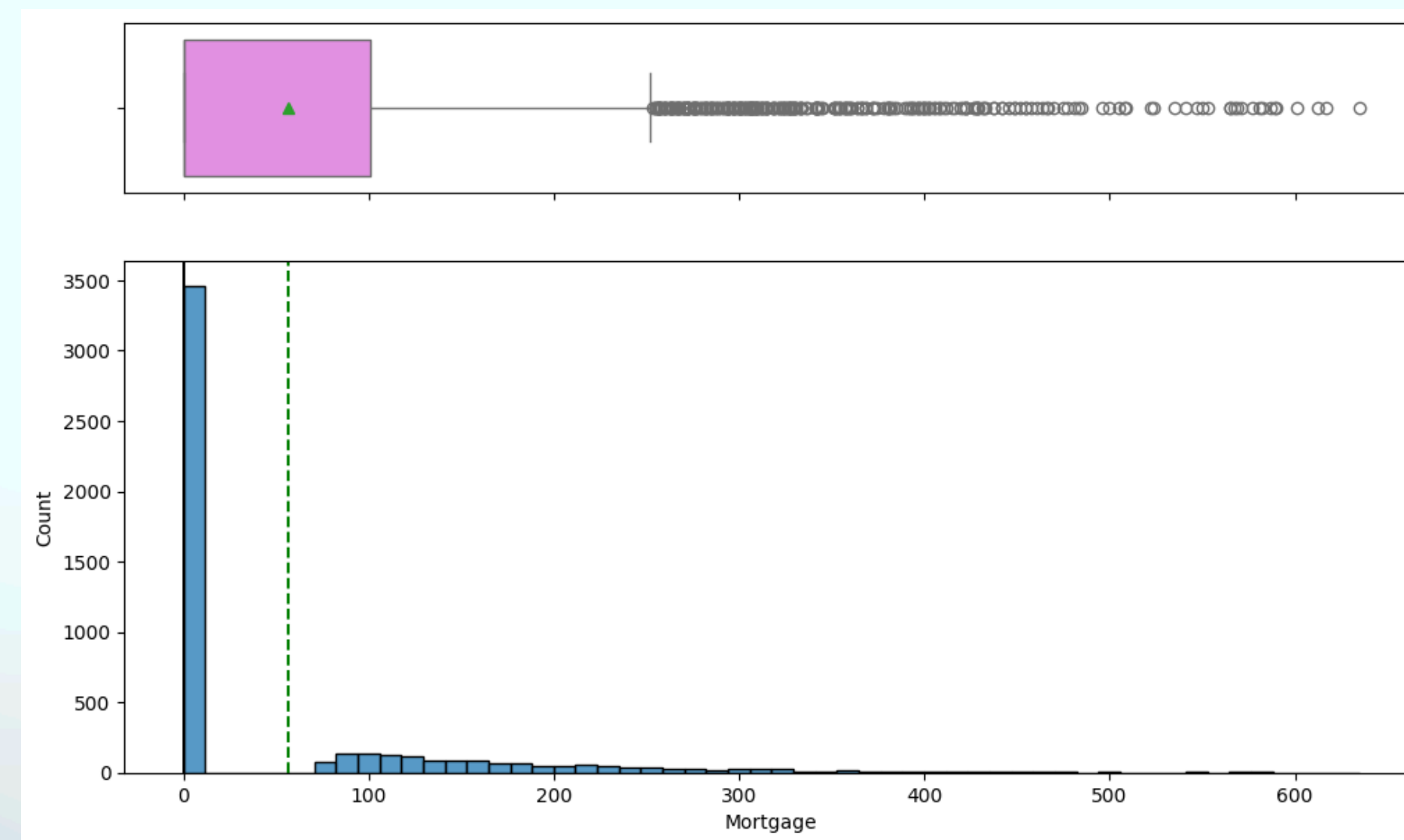
Business problem Overview and Solution Approach

- The problem we have here is AllLife Bank has a business model that relies on growing its loan portfolio to drive sustainability through interest income
 - They rely currently on a large base of depositors rather than borrowers and they need help converting the depositors to borrowers
- A solution to accomplish this goal will be to build a predictive model to predict which customers would be great candidates for the different types of loans

EDA Results - Question 1

Distribution of Mortgage Attributes

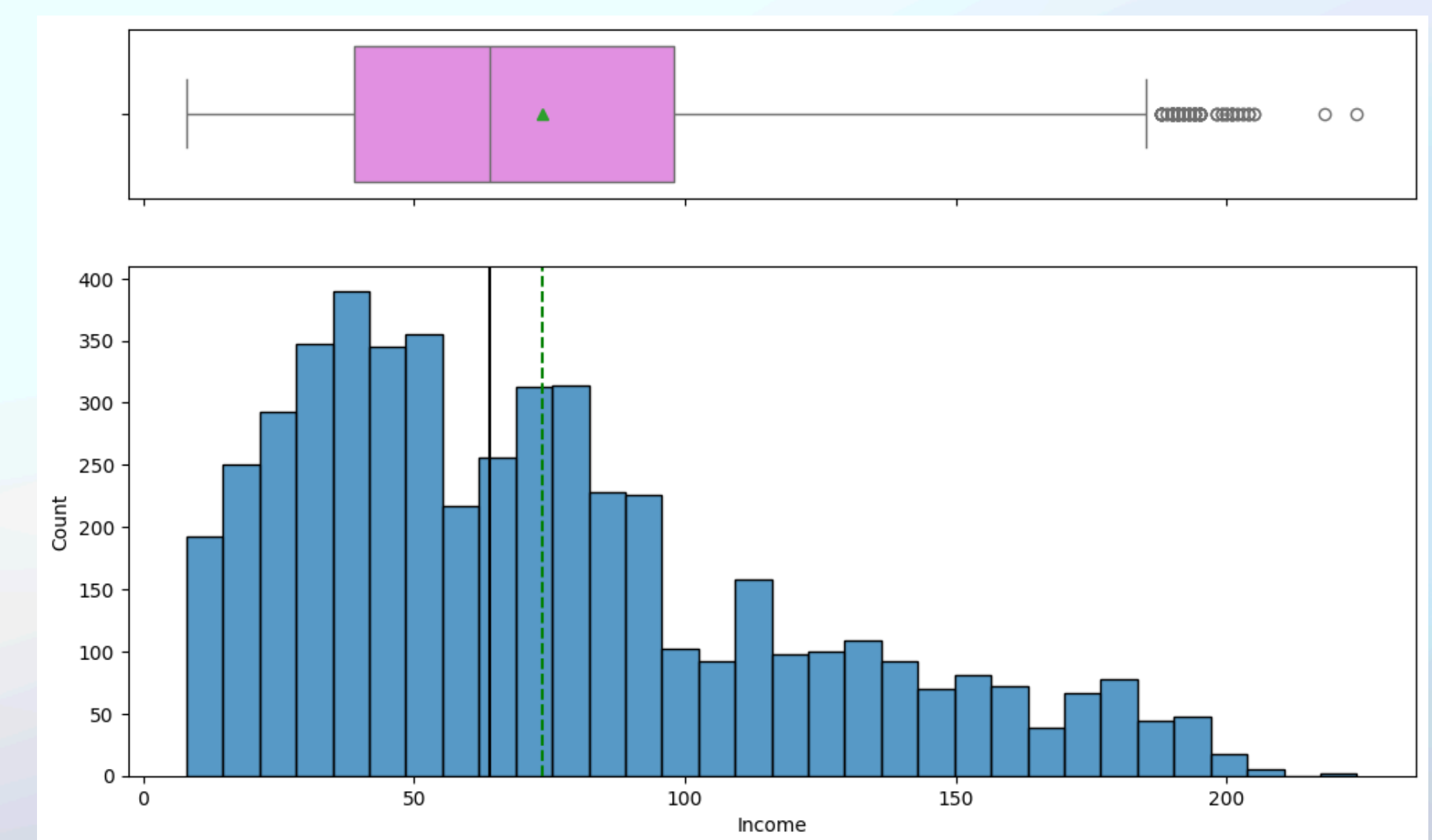
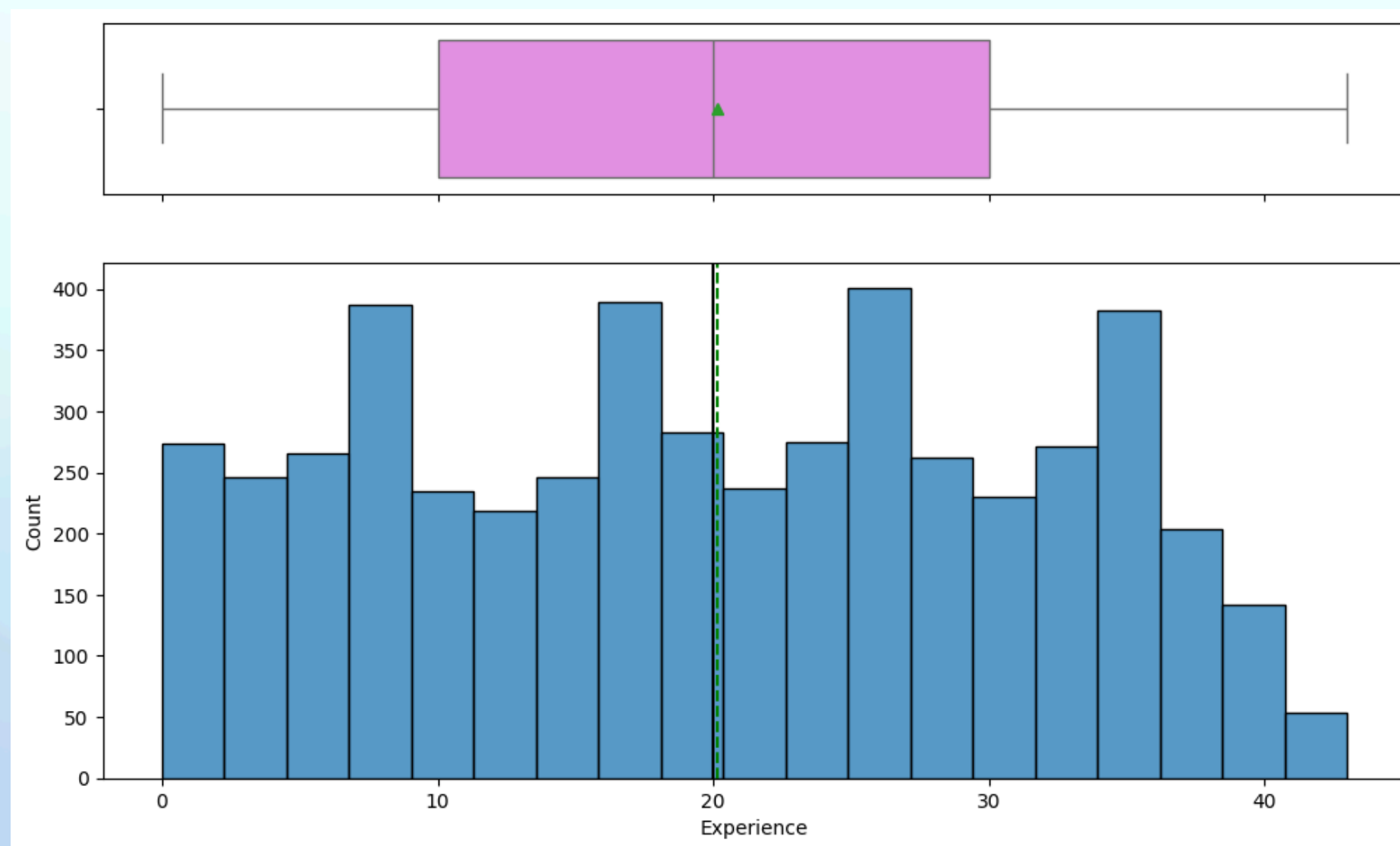
We can see that majority of the customers do not have a mortgage. There are some outliers which are between \$300,000 - \$650,000 and the mean is about \$70,000



EDA Attributes - Question 1 Cont.

Observations on Experience & Income

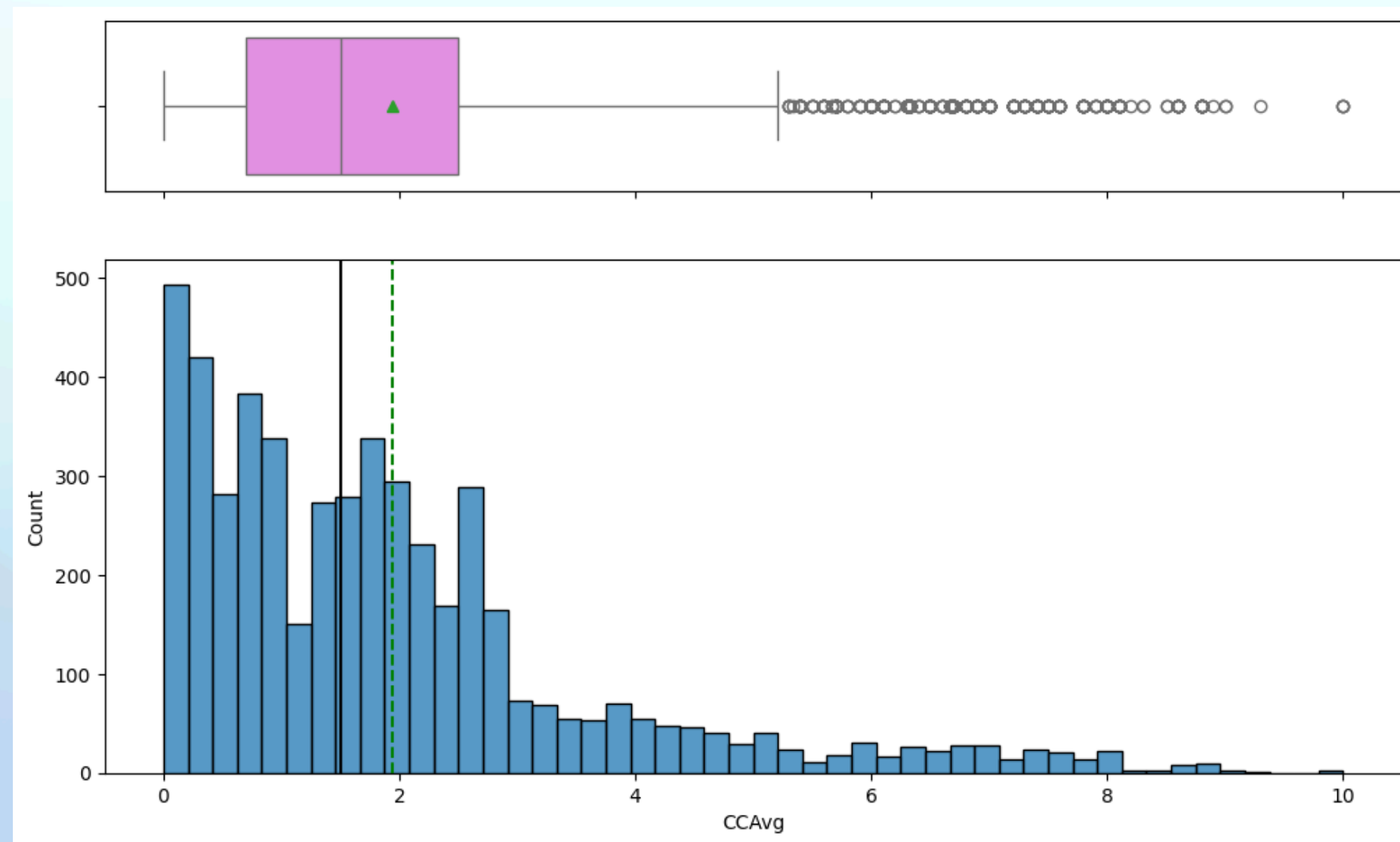
Here we see the mean of the years of professional experience is ~20 years and the mean income for our customers is ~\$73,000. We see the experience range from 0 - 43 years and. Income from \$8,000 - \$224,000



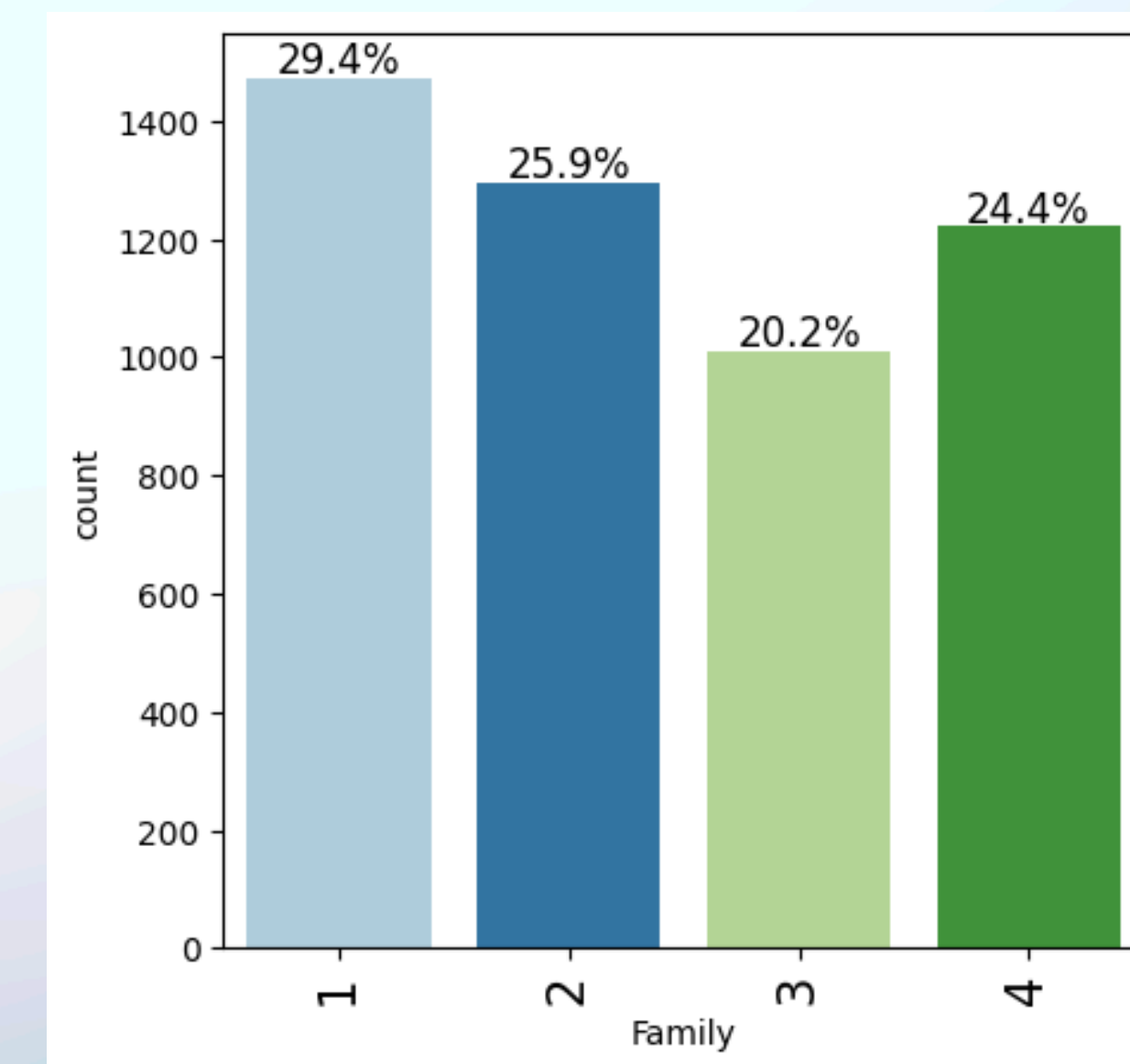
EDA Attributes - Question 1 Cont.

Observations on CCAvg & Family

We can see that the mean spending in months is ~\$ 1,930 with the max spending reaching \$10,000. There are many outliers from ~\$5,000 - \$10,000.



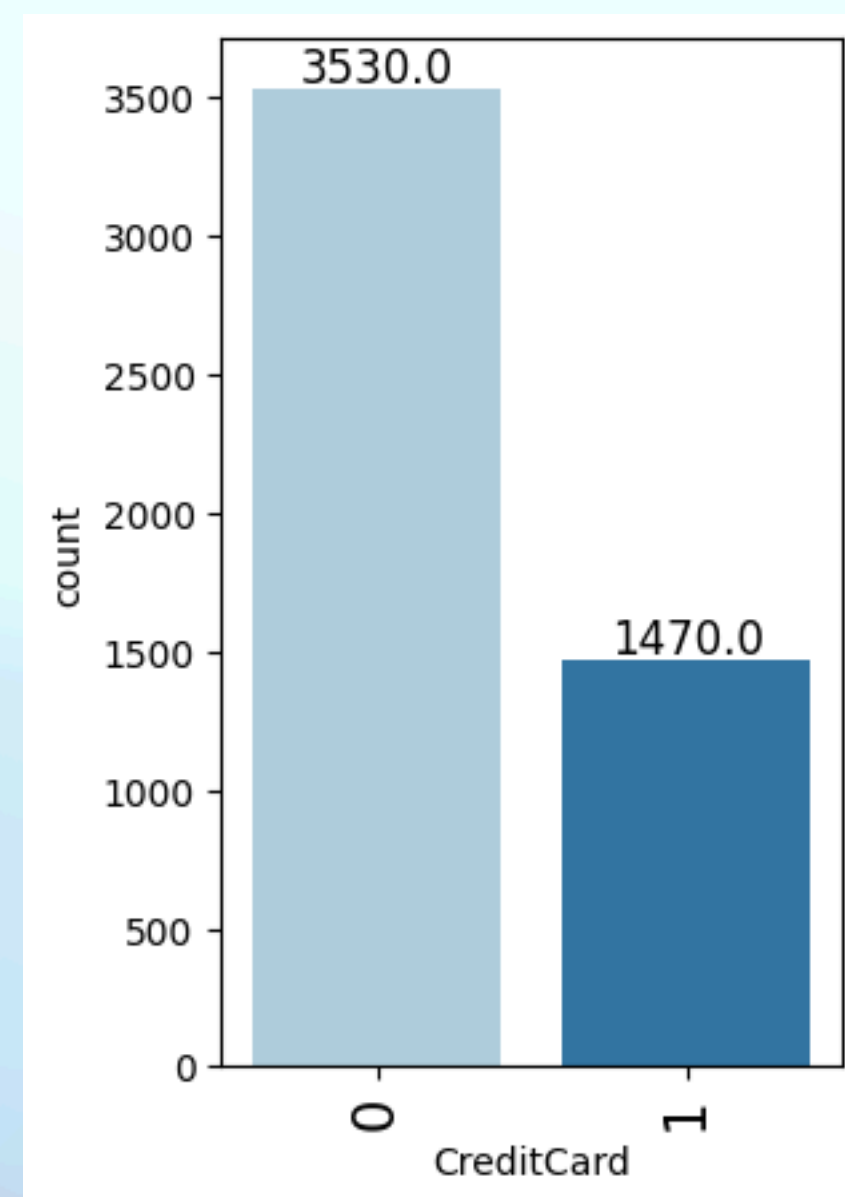
We see the mean family size is 2.39 with the Minimum family size being 1 person and the maximum family size being 4 people.



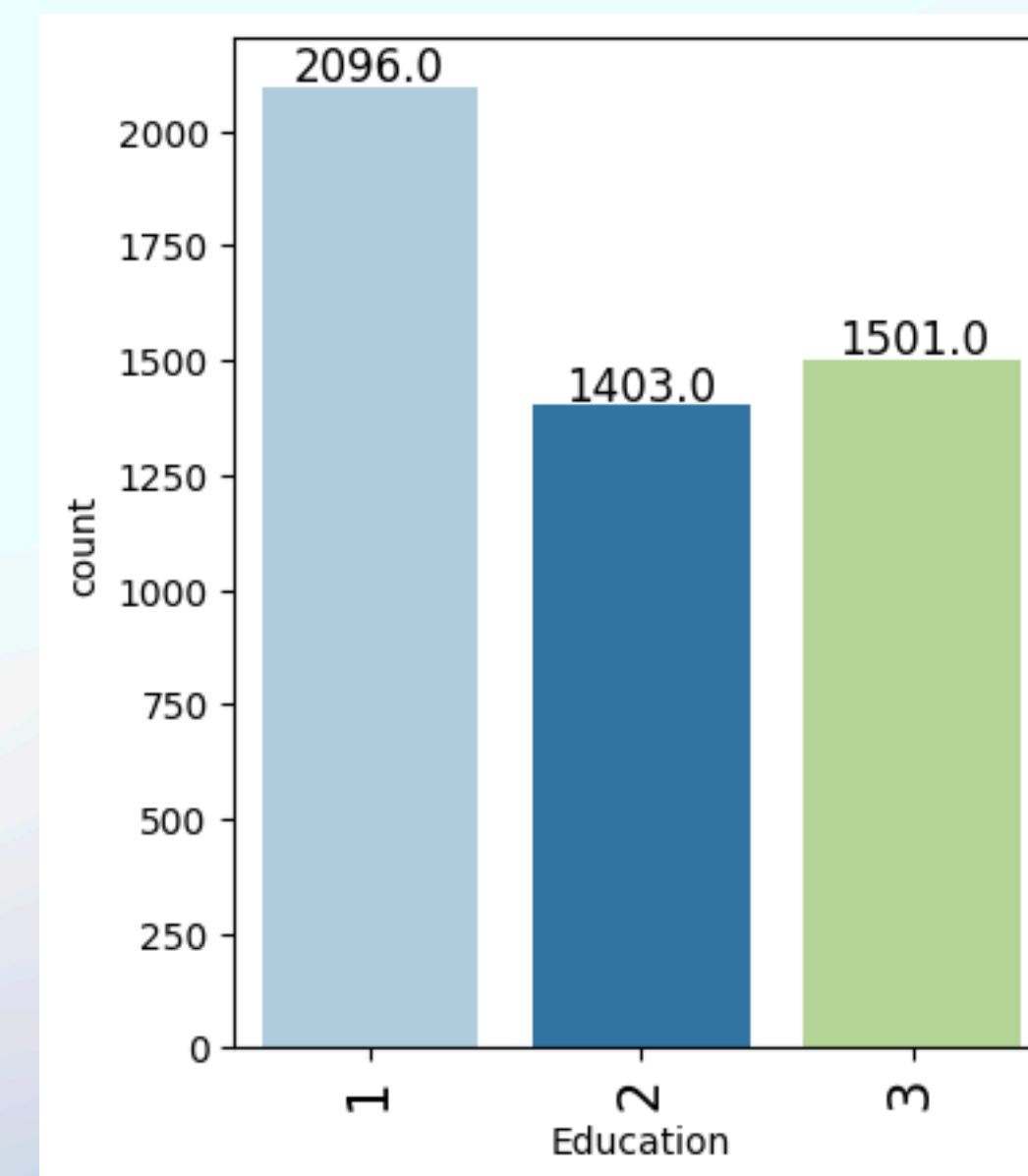
EDA Attributes - Question 2

Observations on Credit Cards and Education

We see that the majority of the customers do not have a credit card. 3,530 to be exact. 1,470 of the customers have a credit card.



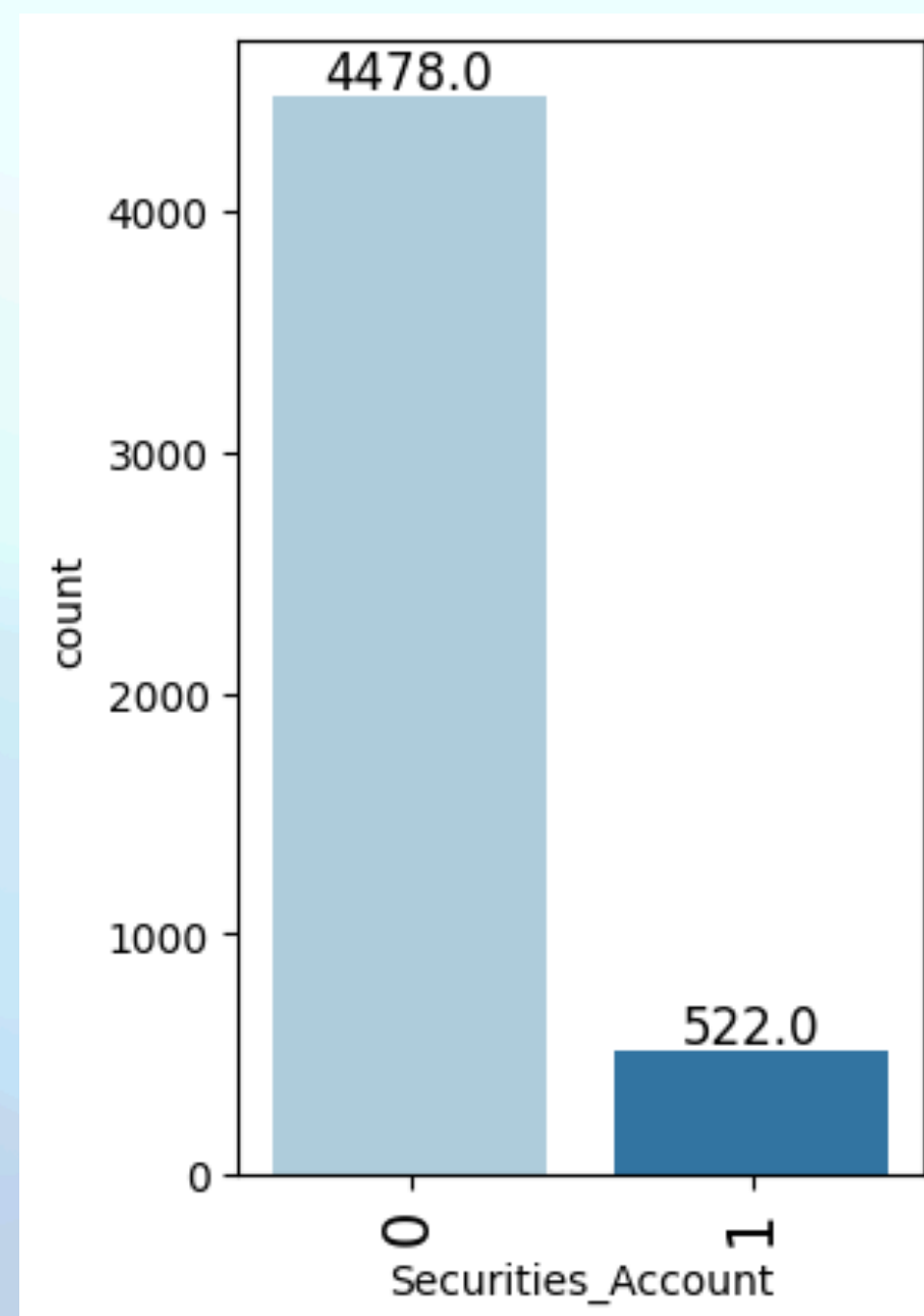
We see that 2,096 customers have an undergraduate degree, 1,403 have a masters degree and 1,501 have an advanced/professional degree.



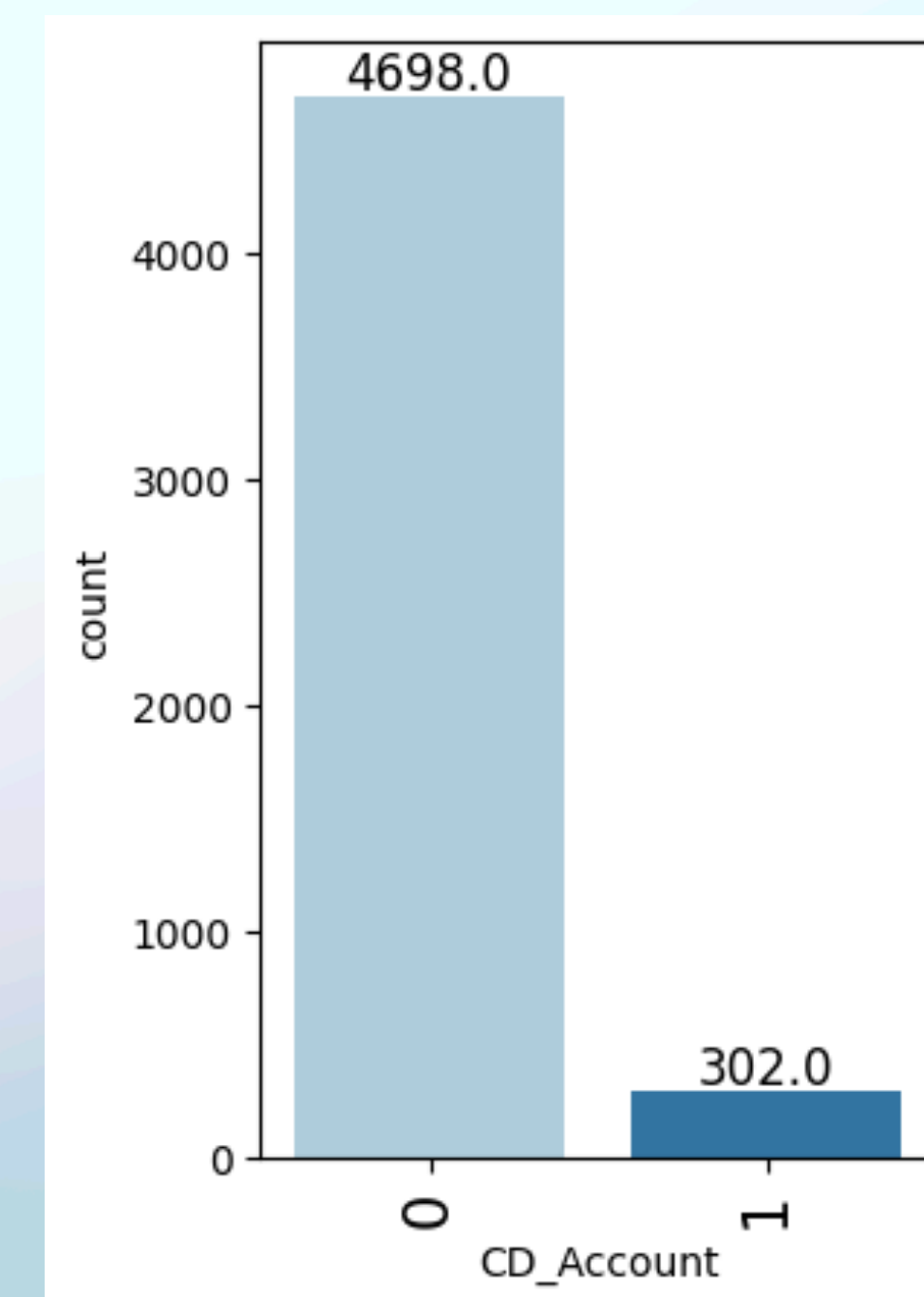
EDA Attributes - Question 2 Cont.

Observations on Securities Account & CD Account

We see that the 4,478 customers do not have a security account and 522 do have a security account.



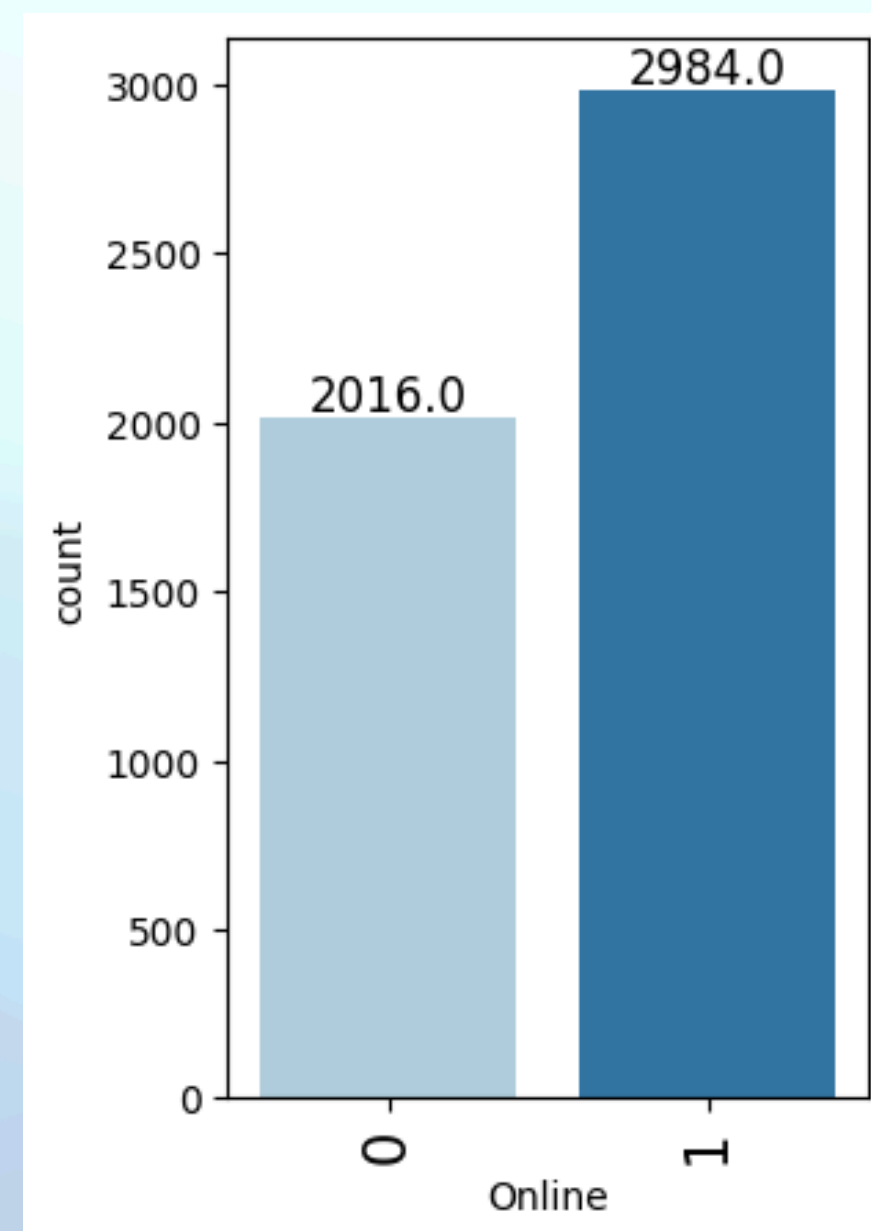
We see that 4,698 do not have a certificate of deposit account and 302 customers do have a certificate of deposit account.



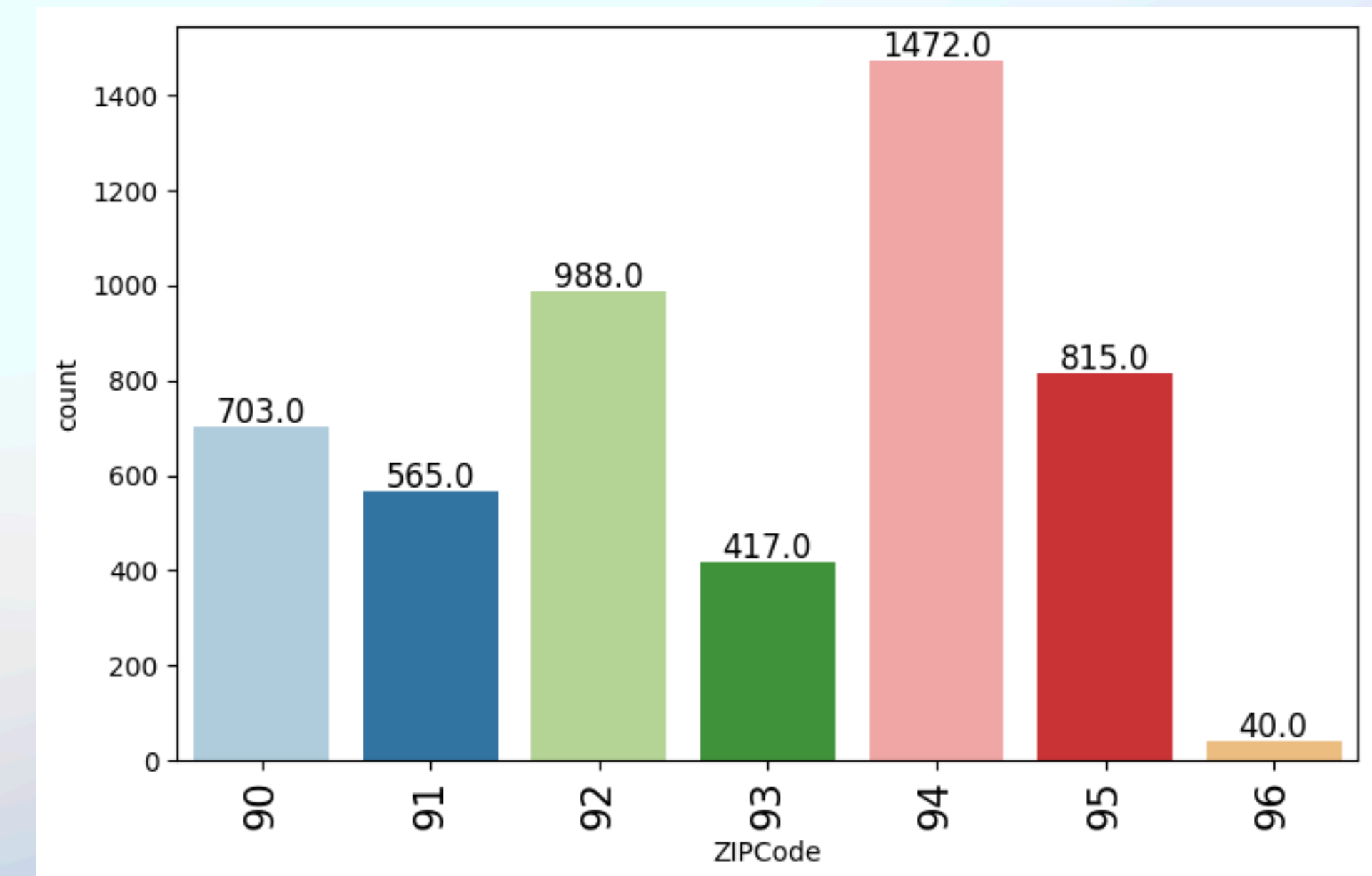
EDA Attributes - Question 2 Cont.

Observations on Online Users & Zip Codes

We see that the 2,016 customers do not use internet banking facilities and 2,984 do use Internet banking facilities.



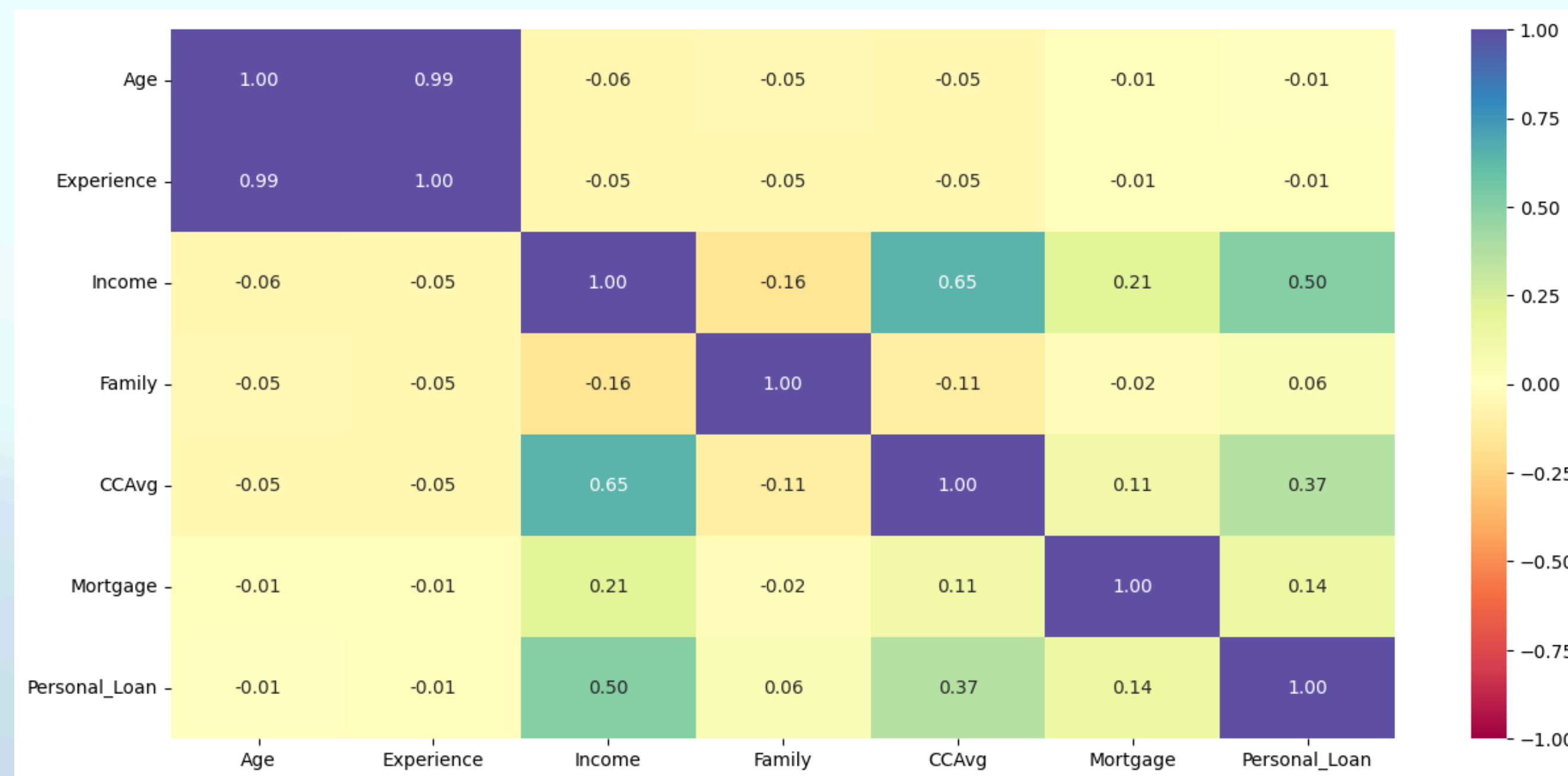
There are too many zip codes to graph so we group them by the first two numbers. We see that the zip codes with the highest number of customers starts with '94' and the zip codes with the lowest number of customers starts with '96'.



EDA Attributes - Question 3

Observations on correlation of all attributes

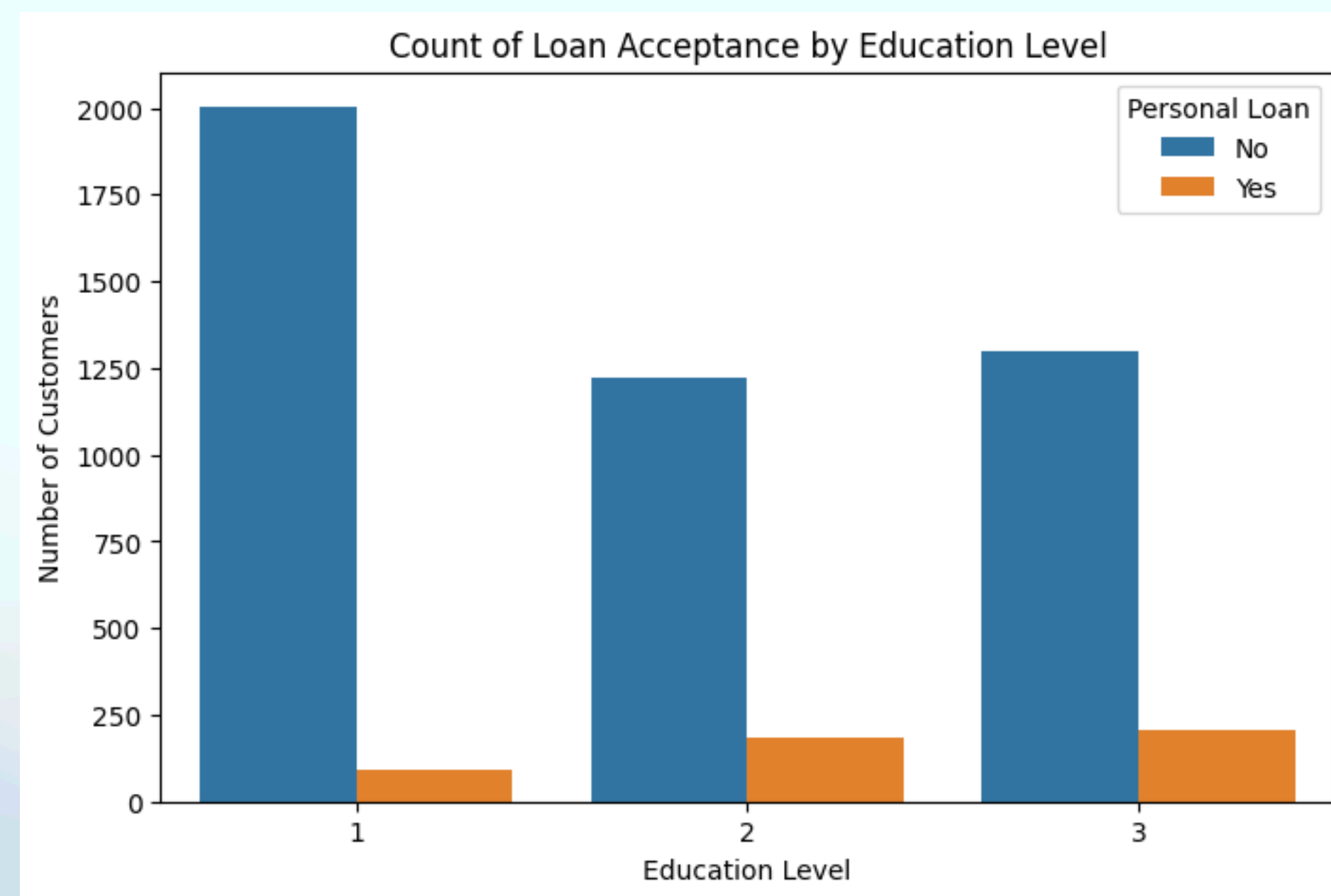
The attribute that has the strongest positive correlation with 'Personal_Loan' is 'Income'. The attributes with the most negative correlation to 'Personal_Loan' are 'Age' and 'Experience'.



EDA Attributes - Question 4

Customer's interest in purchasing a loan vs their education

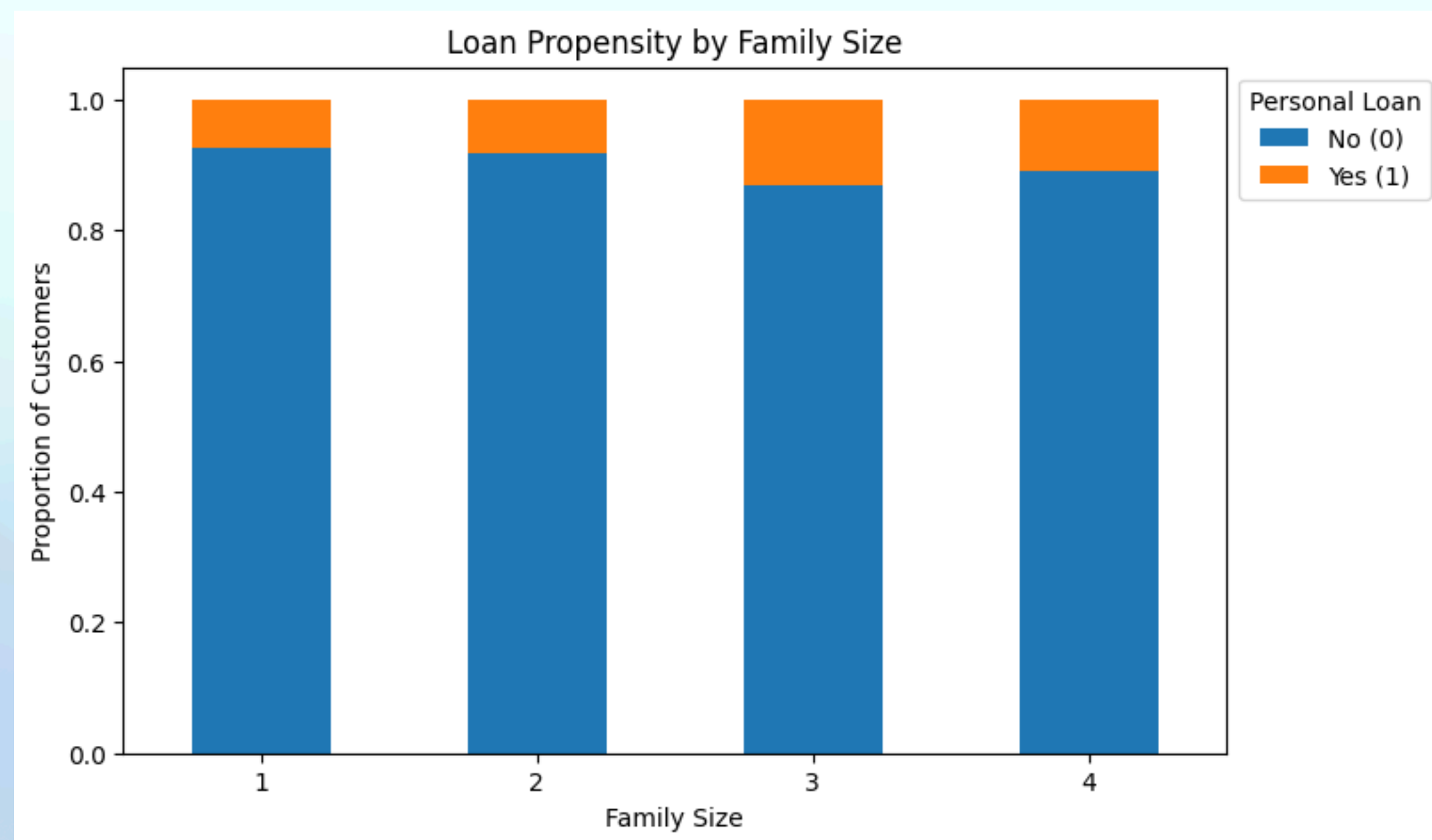
We see that the higher the education level the more interest they have in purchasing a loan. There are about 2,500 customers that have education level 2 or 3.



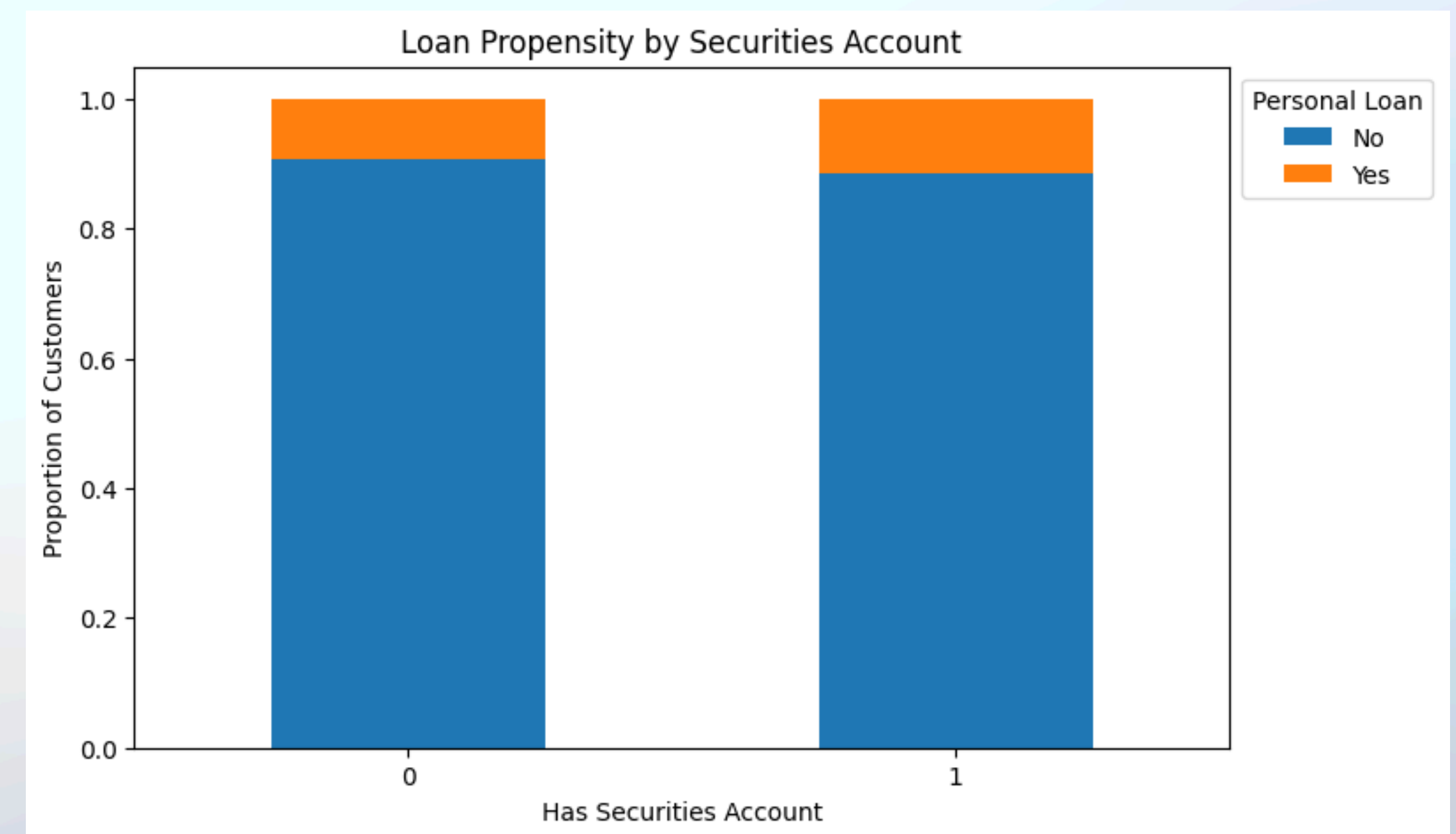
EDA Attributes - Question 4 Cont.

Observations on Personal Loan vs Families & Securities Account

We see that the majority of the customers do not take out a personal loan. Typically the bigger the family the more likely they take out a personal loan.



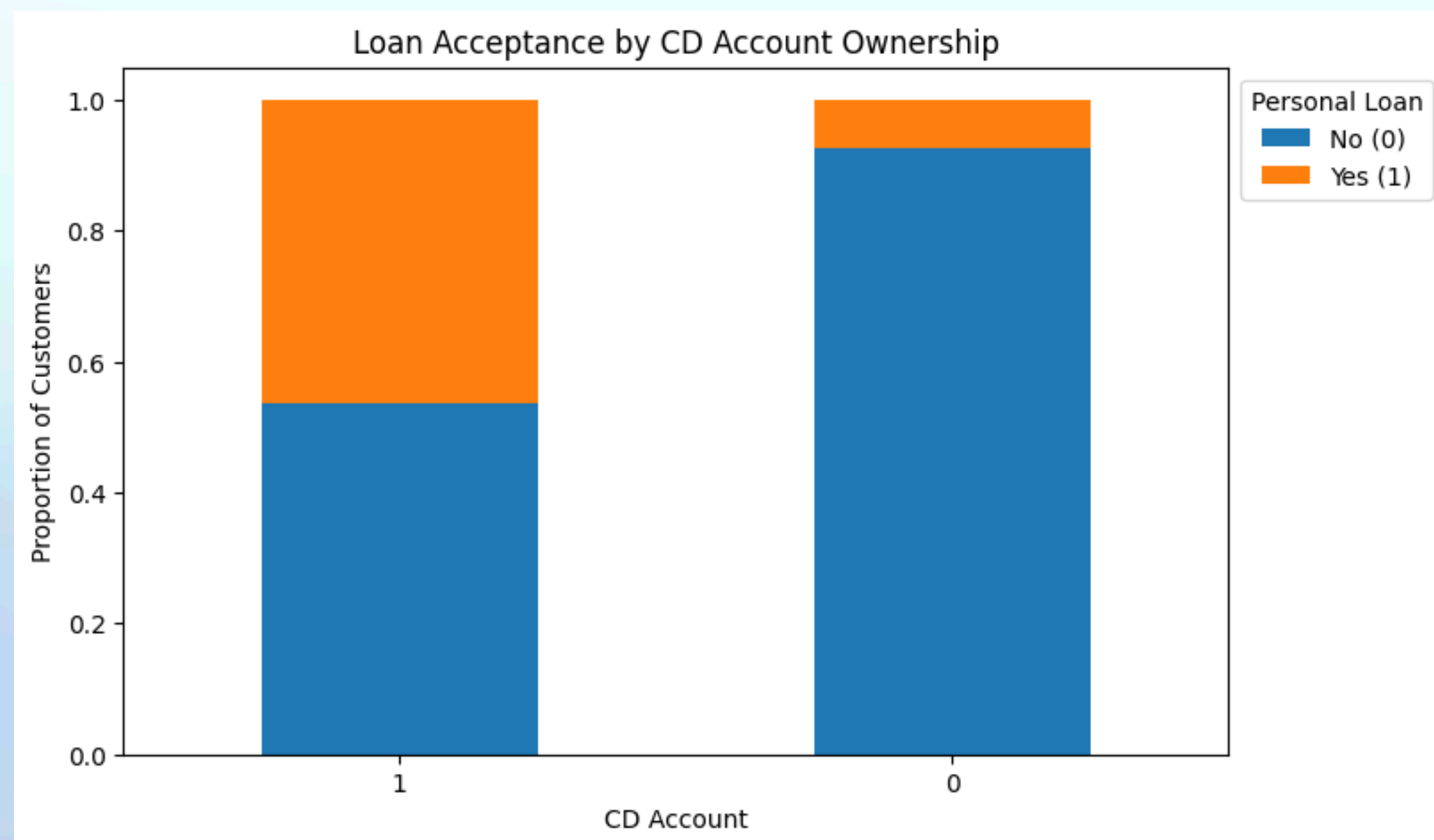
We see that customer who do have Securities Account are more likely to take out a personal loan than someone who does not have one.



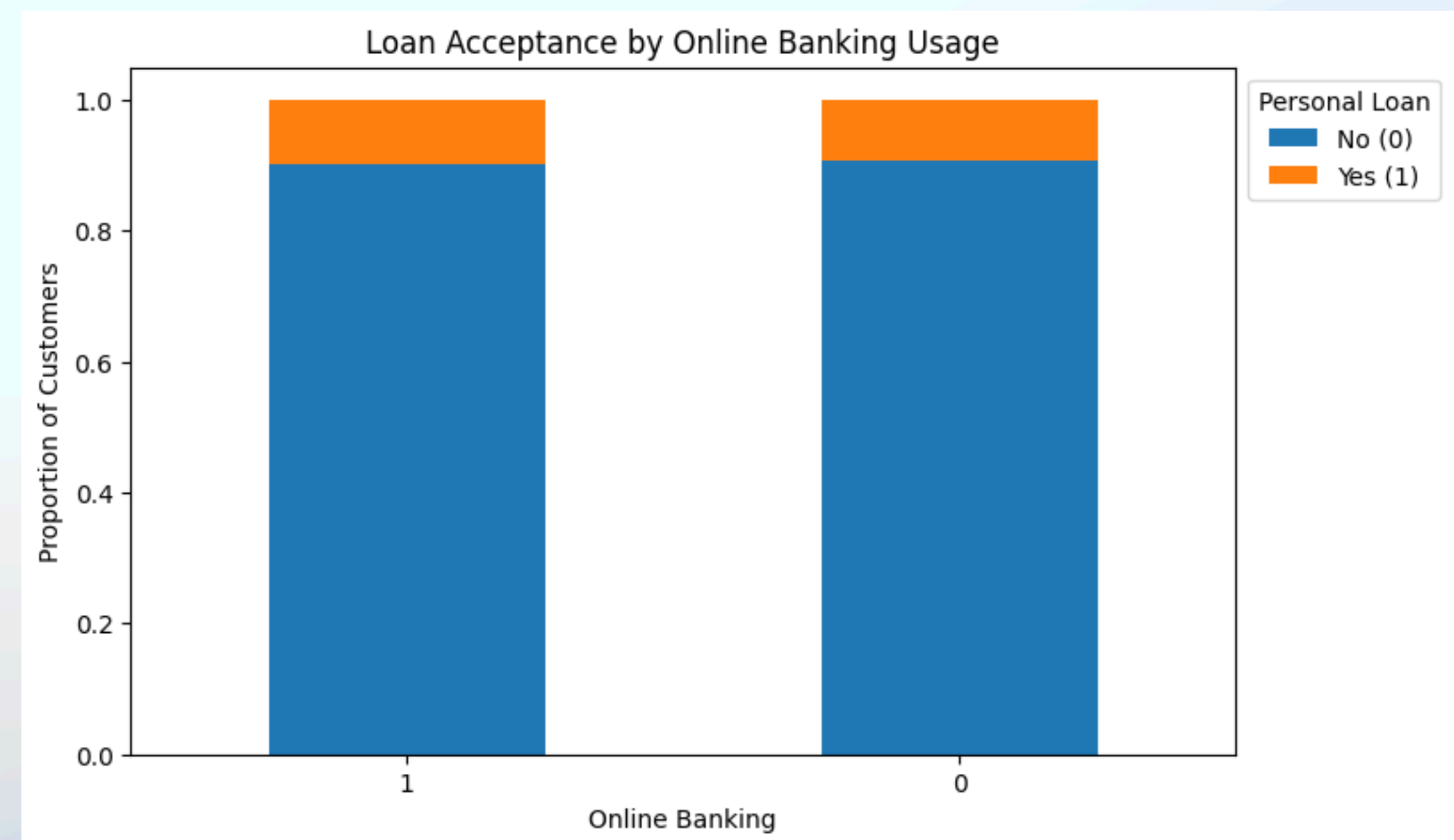
EDA Attributes - Question 4 Cont.

Observations on Personal Loan vs CD Account & Online

We see that the majority of the customers that do have a CD Account are more interested in taking out a personal loan than those who do not have a CD Account.



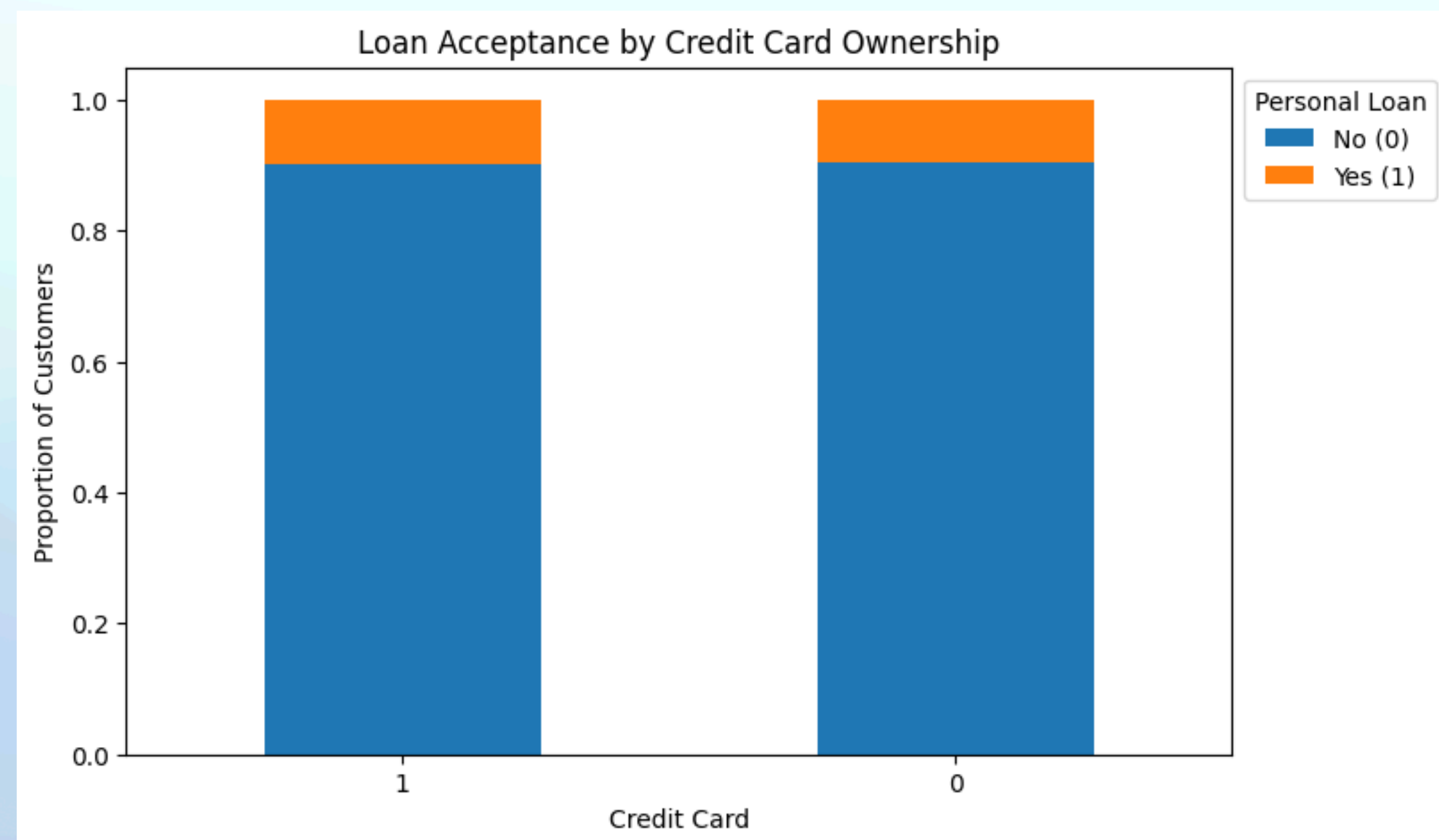
We see that customer who do use online banking are slightly more likely to take out a personal loan than those who do not use online banking.



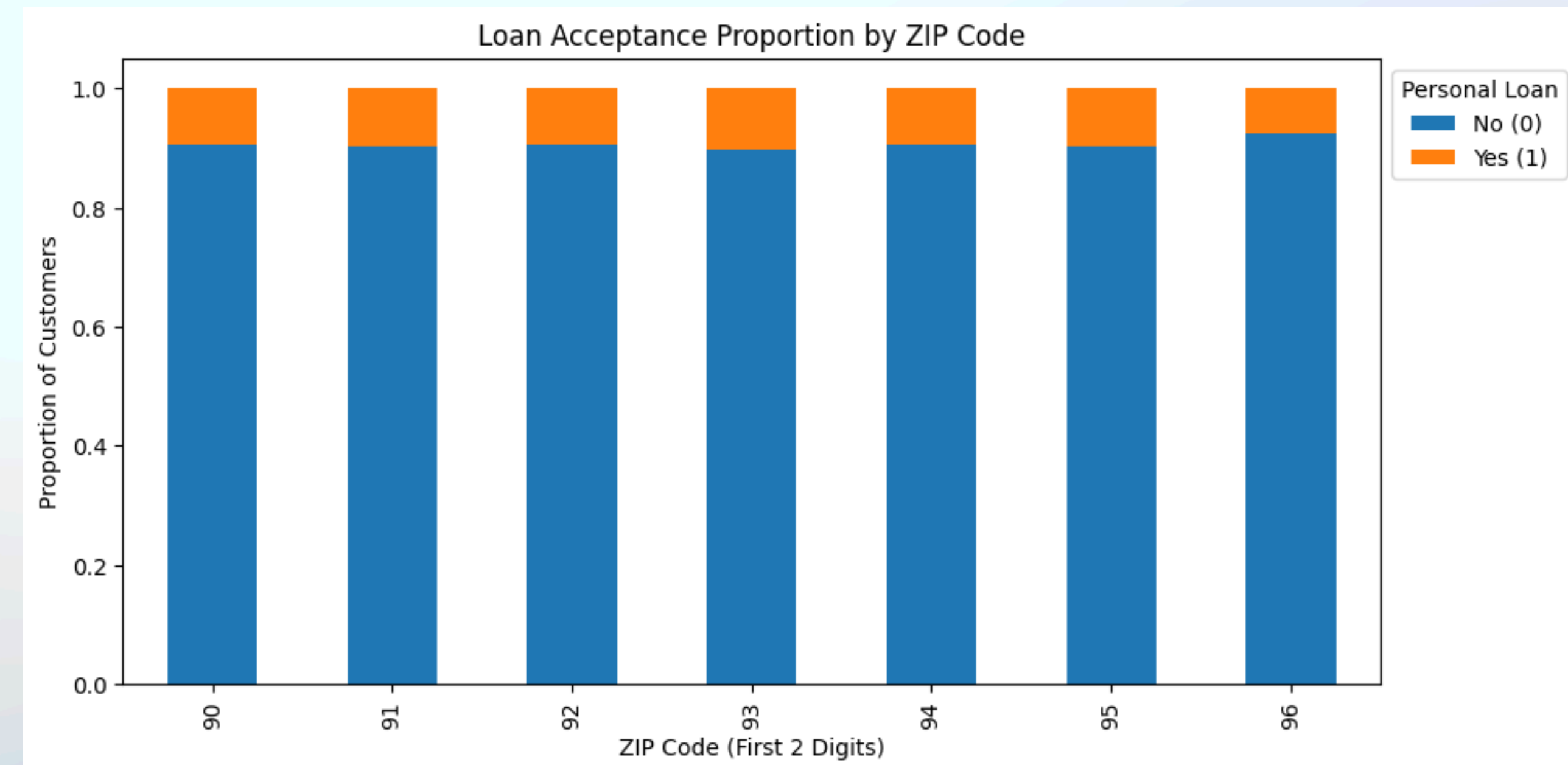
EDA Attributes - Question 4 Cont.

Observations on Personal Loan vs Credit Card & Zip Code

We see that the customers who have a credit card are slightly more likely to get a personal loan.



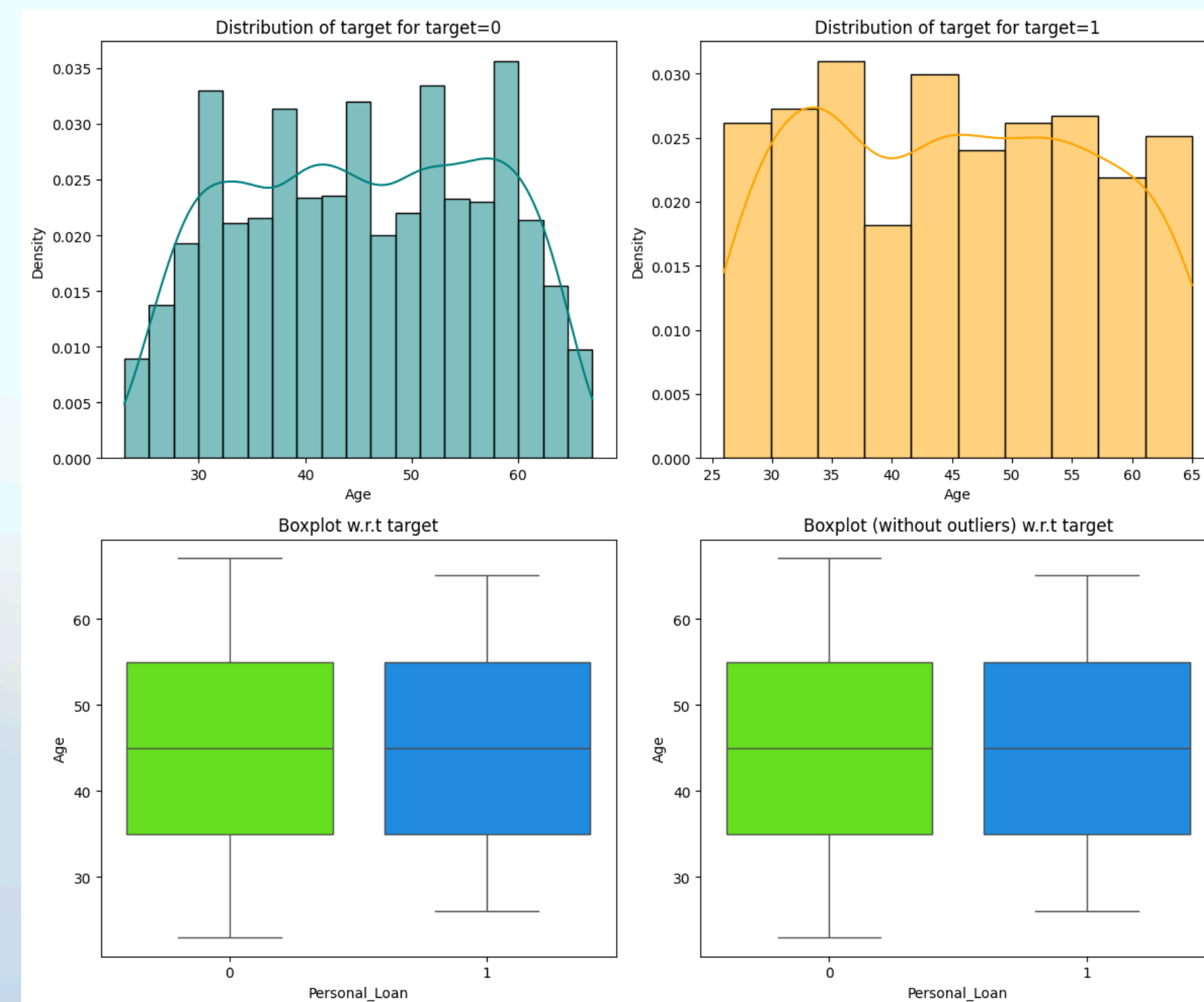
We see that customers in all zip codes are similar in terms of having interest in taking out a personal loan. Zip codes starting in '93' have the most customers who are interested in taking out a personal loan.



EDA Attributes - Question 5

Observations on Personal Loan vs Age

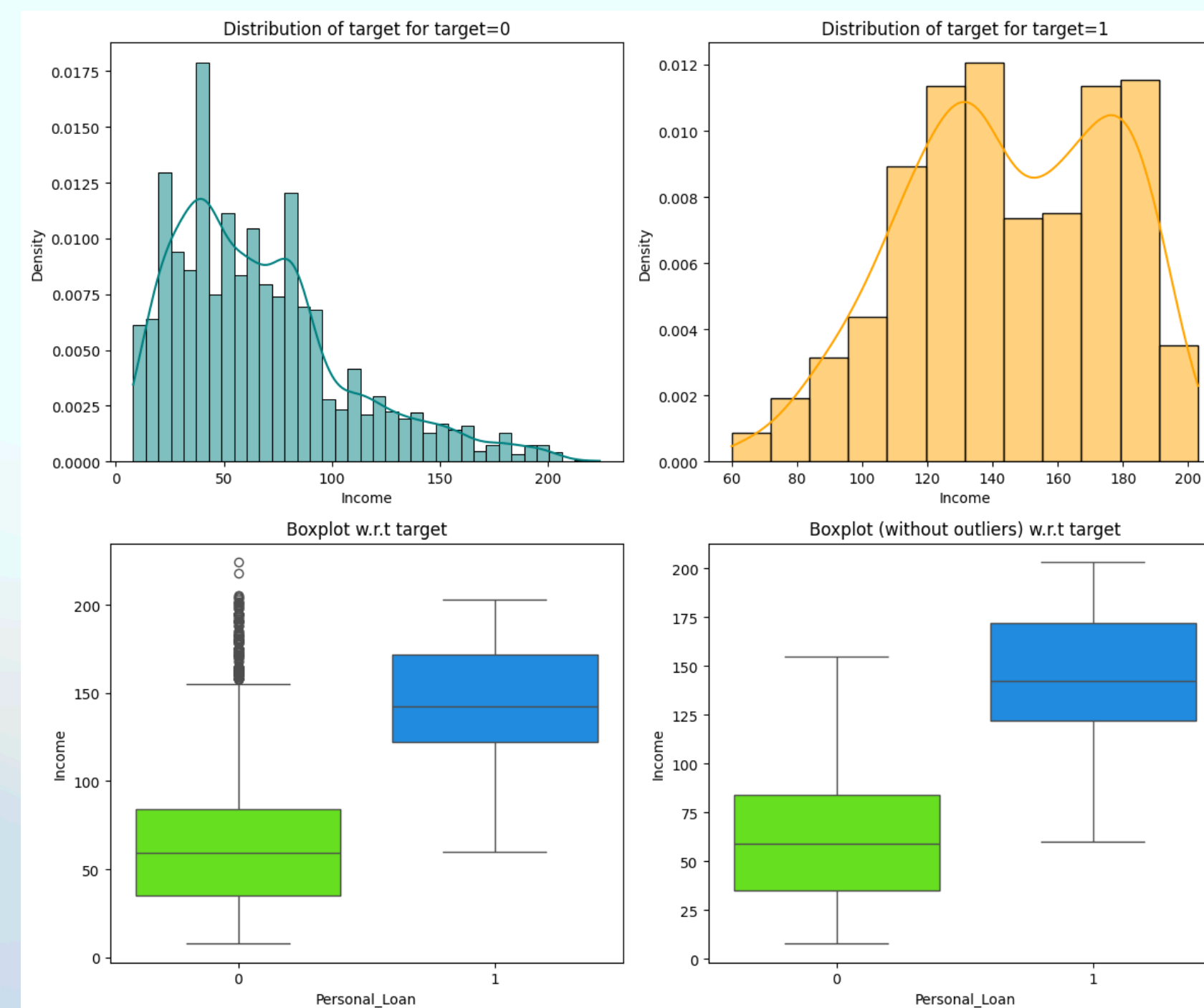
We see that the customers who take out a loan are mostly evenly distributed across the 25-65 age range. The Median age for both groups who did and did not take out a loan is at 45 years old.



EDA Attributes - Question 5 Cont.

Observations on Personal Loan vs Income

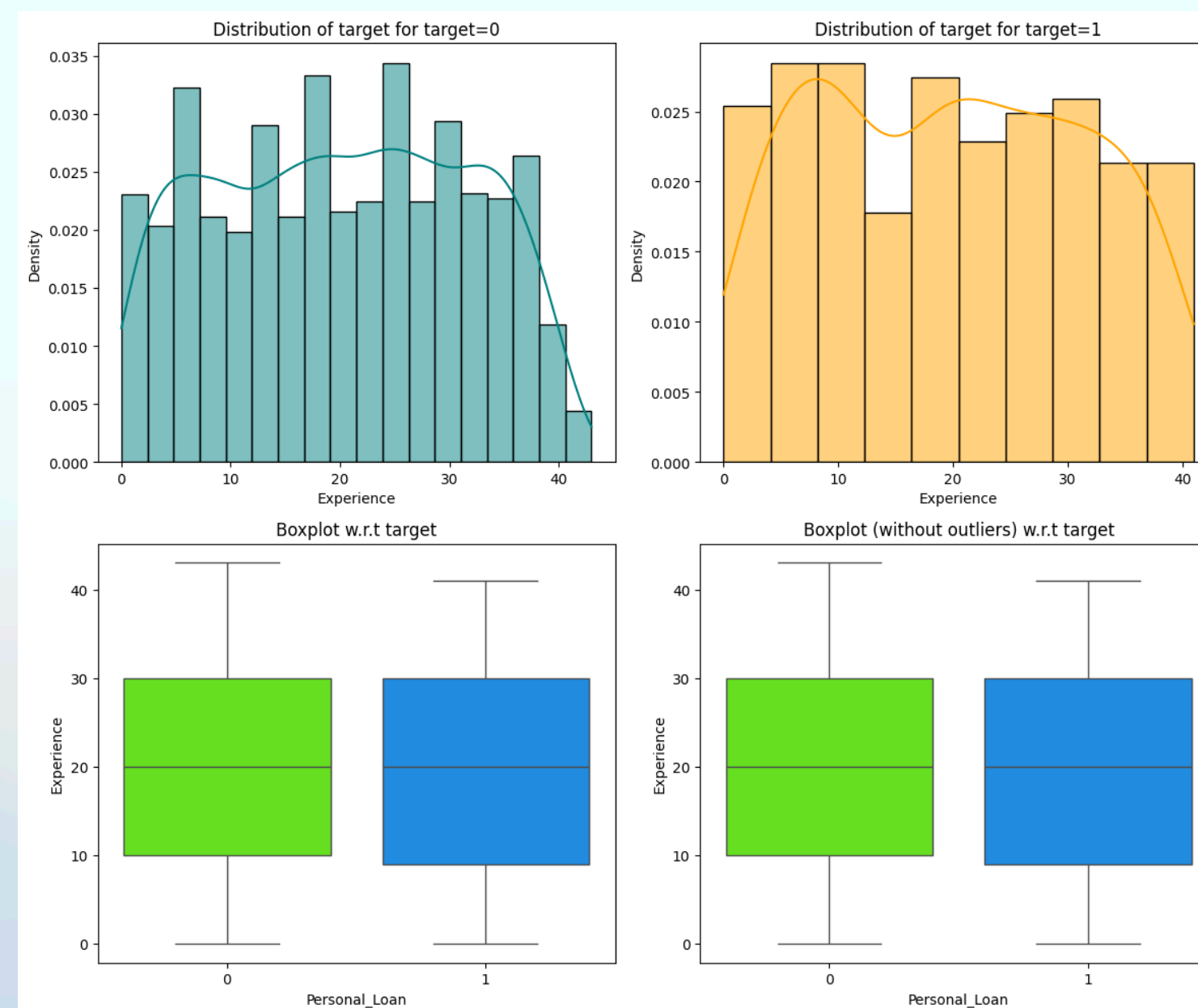
We see that the customers who are non borrowers are mostly earning around \$30,000 - \$80,000 and as income increases, the number of depositors drops. We see in the group of borrowers the higher the income the more likely they take out a personal loan. The majority of the borrowers earn ~\$130,000 - ~\$180,000.



EDA Attributes - Question 5 Cont.

Observations on Personal Loan vs Experience

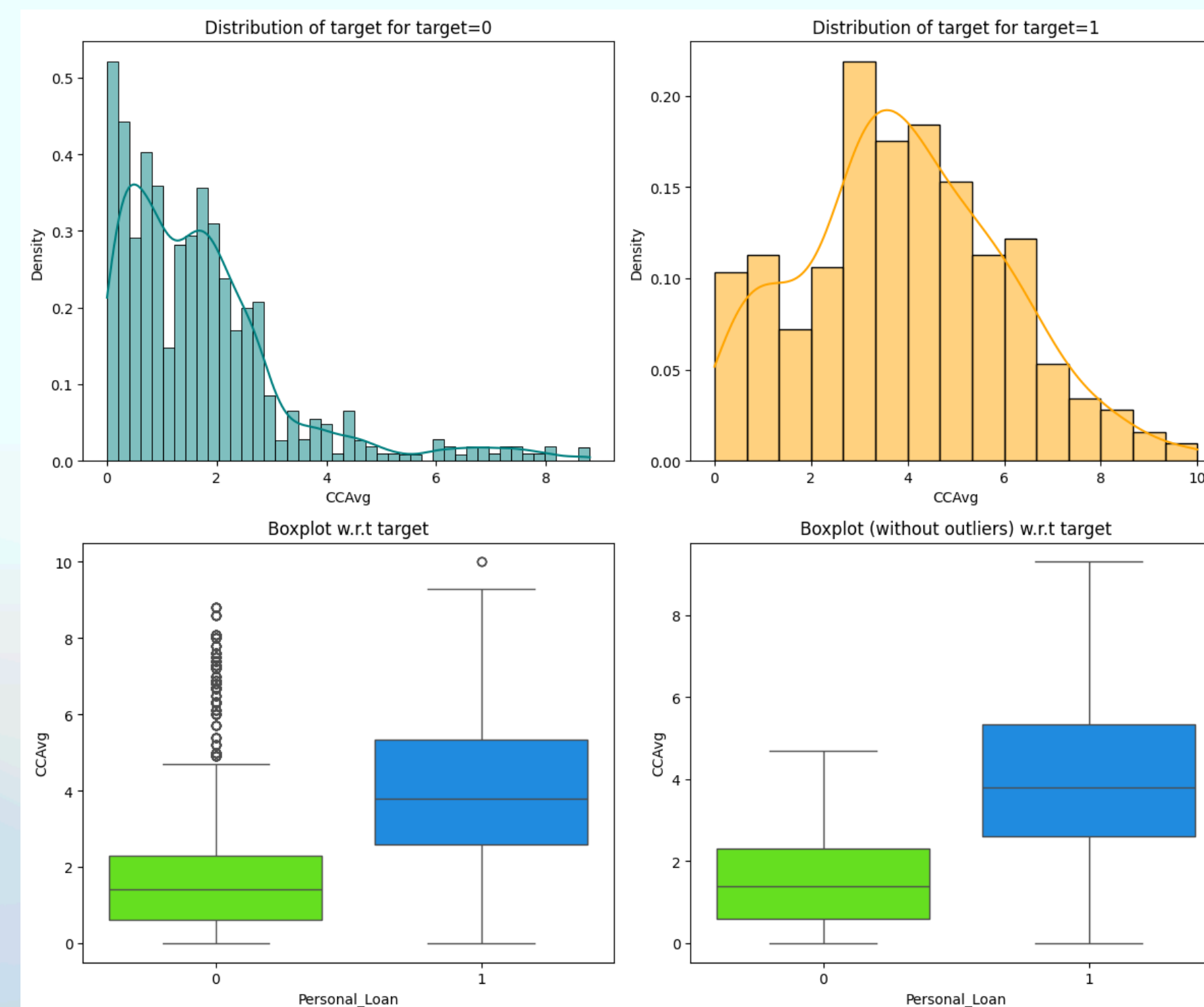
We see that the customers who have taken out a personal loan are mostly evenly distributed with years of experience. The same goes more customers who have not taken out a personal loan.



EDA Attributes - Question 5 Cont.

Observations on Personal Loan vs CCAvg

We see that the customers who do not take out personal loans open very little on their credit cards. Majority of them are spending between \$0 - \$3,000 per month. Customers who do take out personal loans are spending about \$3,000 - \$6,000 per month.



Data Preprocessing

Outliers

- We checked the data to see what outliers there are and found 3 variables with major outliers
 - CCAvg (6.48) is a massive outlier because this high score is telling us this customers monthly credit card spend is like 6+ times the standard deviation above the mean.
 - Mortgage (5.82) is another significant outlier. This person most likely has the highest mortgage.
 - Income (1.92) is another outlier but not as significant as Mortgage or CCAvg.

Data Preprocessing

Data Preparation

- We see that the shapes of the training and test sets are:
 - Training set has 3,500 rows and 17 columns
 - Test set has 1,500 rows and 17 columns
- Percentage of classes in the training and test set are:
 - In the training set 90.54% do not have a personal loan and 9.45% do have a personal loan
 - In the test set 90.06% do not have a personal loan and 9.93% so do have a personal loan
- Both the training and test sets are consisted of float values
- There are no duplicate or missing values

Model Building

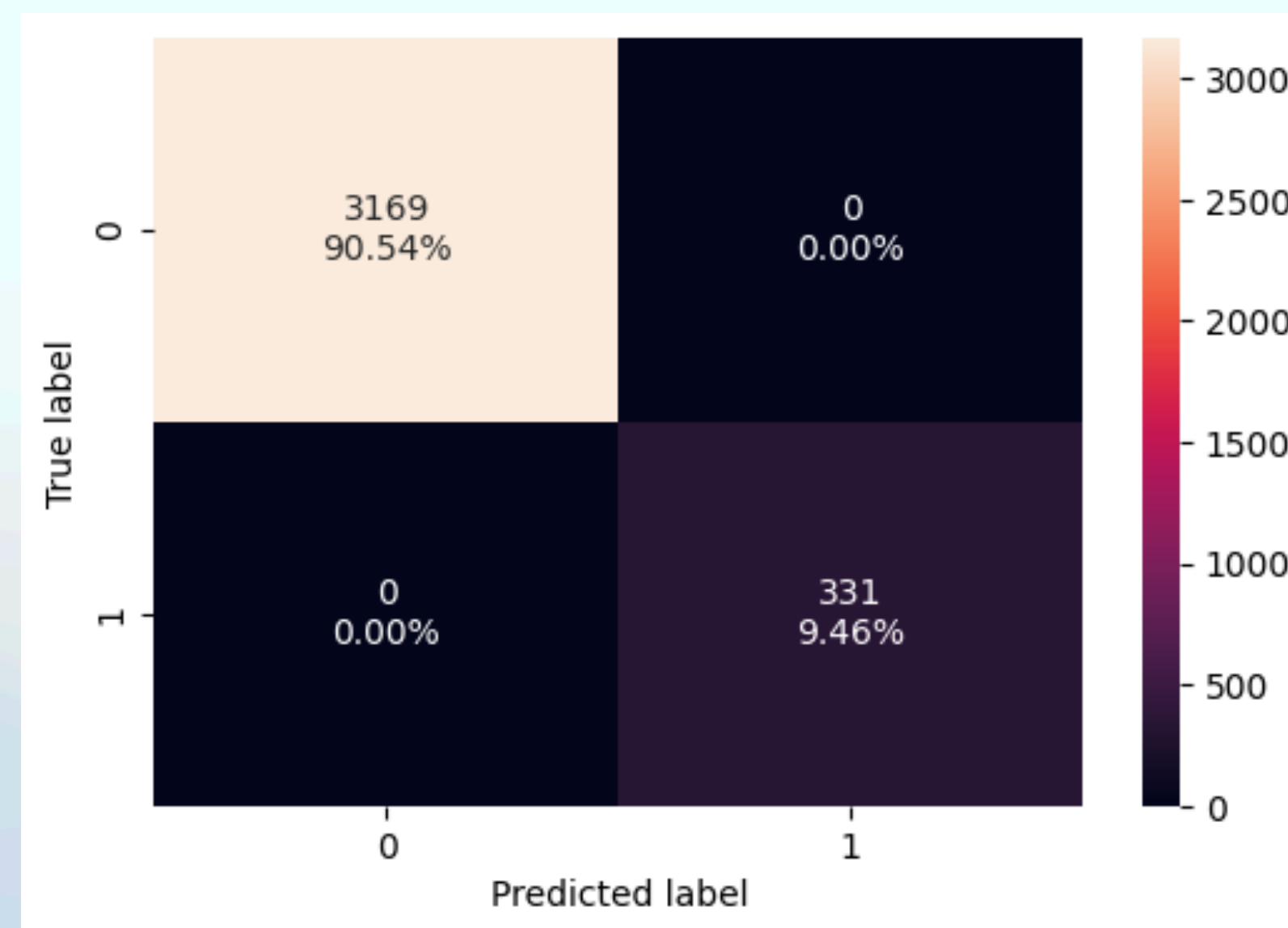
Model Building Steps

- We need to create a Decision Tree and we will do that using these 4 metrics:
 - Accuracy: to see how many total customers did the bank guess right
 - Recall: to see if the bank found all the potential borrowers
 - Precision: tells us how reliable the prediction of someone saying yes is
 - F1-Score - we need this to see how accurately our model is performing. For example if our model is predicting a customer will take out a loan and is right 100% of the time this is a good model. If it finds all the customers who want a loan then it is a perfect model.

Model Building

Checking Model Performance on Training Data

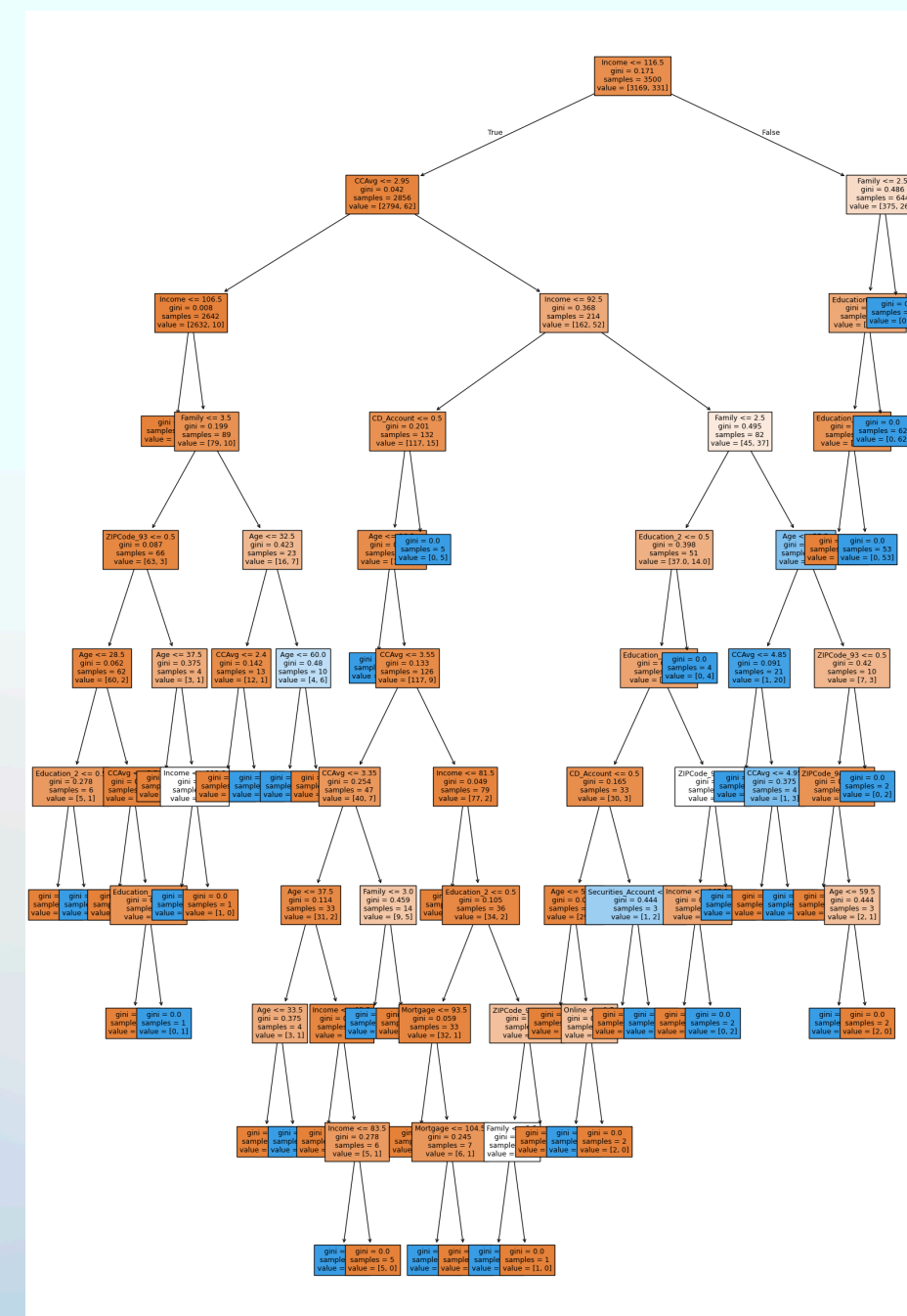
We see that the model is perfect but is more than likely because our Training data is overfitted. The model correctly identified everyone who said no to a loan and every person who said yes to a loan. The model is 'memorizing' every person in the dataset rather than going by rules.



Model Building

Visualizing the Decision Tree

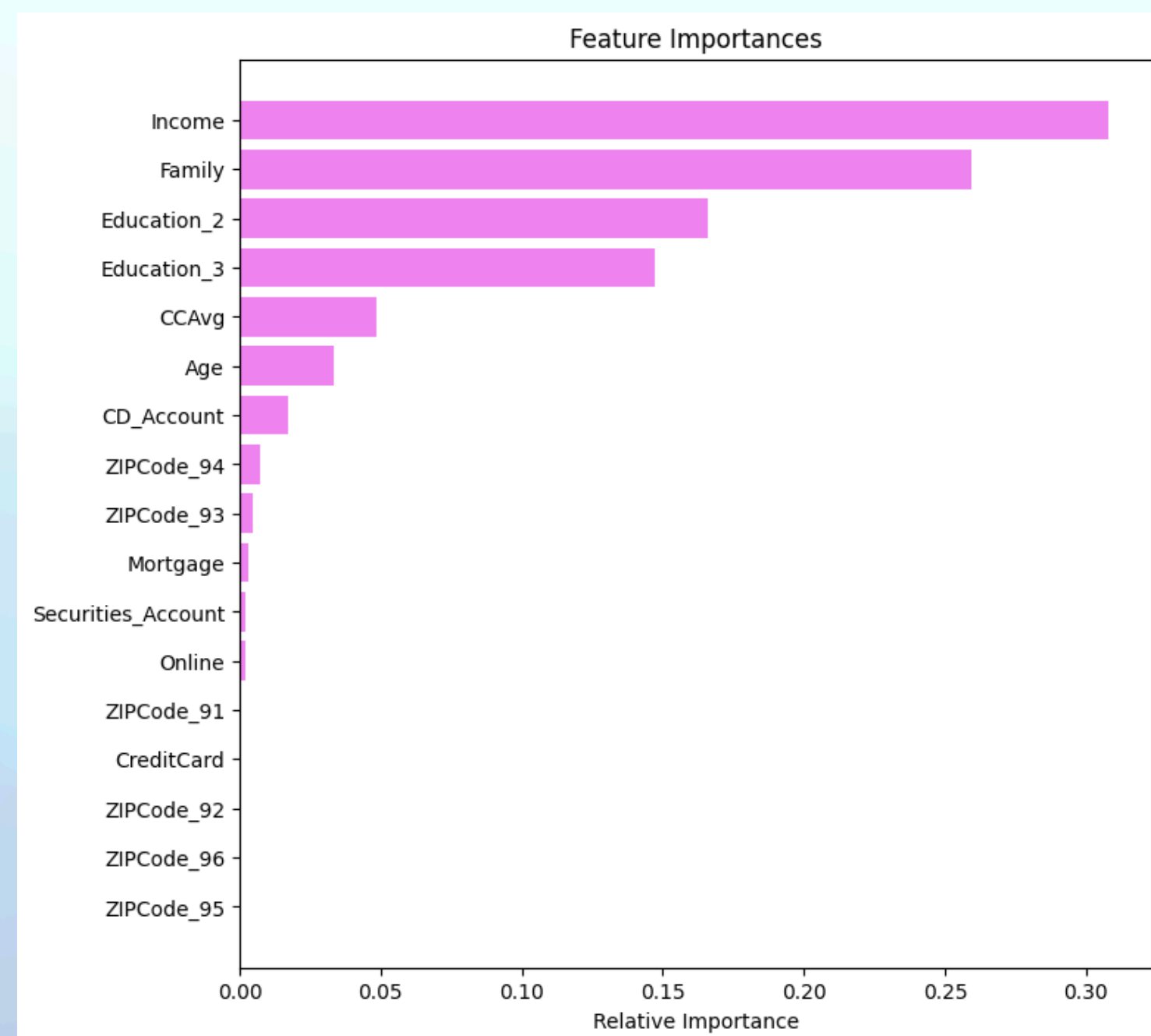
We see that the model's decision tree is overfitted. We know this because the tree depth is way too deep. The decision tree is more focused on finding individual people in the dataset rather than learning which customers take loans. We also see that the tree starts by splitting the dataset in 2 groups based on $\text{income} \leq 110.5$ and then splits it into $\text{CCAvg} \leq 2.95$ then $\text{income} \leq 106.50$.



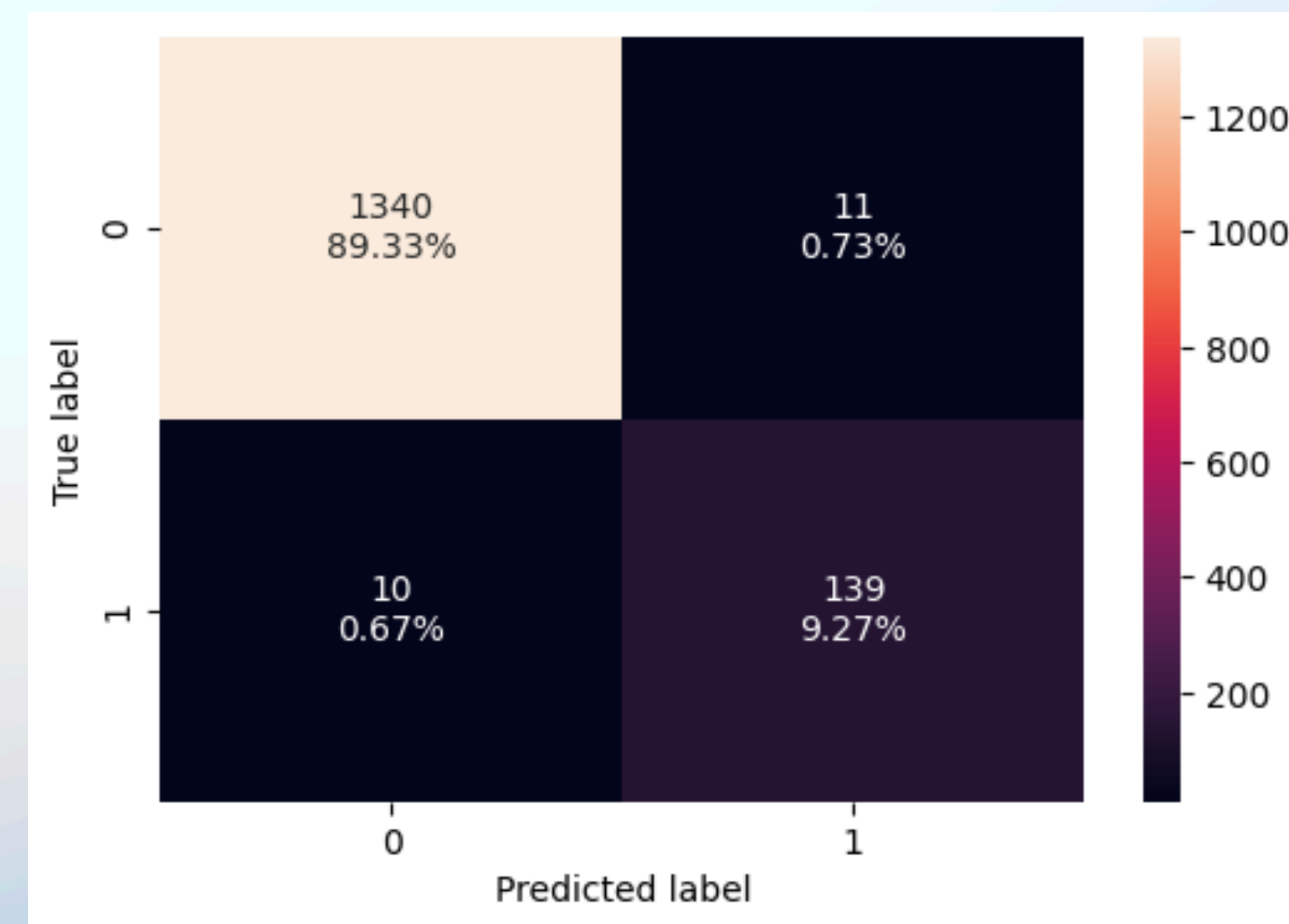
Model Building

Checking Feature Importances and Test Data

We see that the most important features of the model are income, Family, Education_2 & Education_3.



We see that our test data has 11 False Positives and 10 False Negatives. This means our model is using general rules such as income < 110.5 and CCAvg > 2.95 rather than memorizing the customers in the dataset.

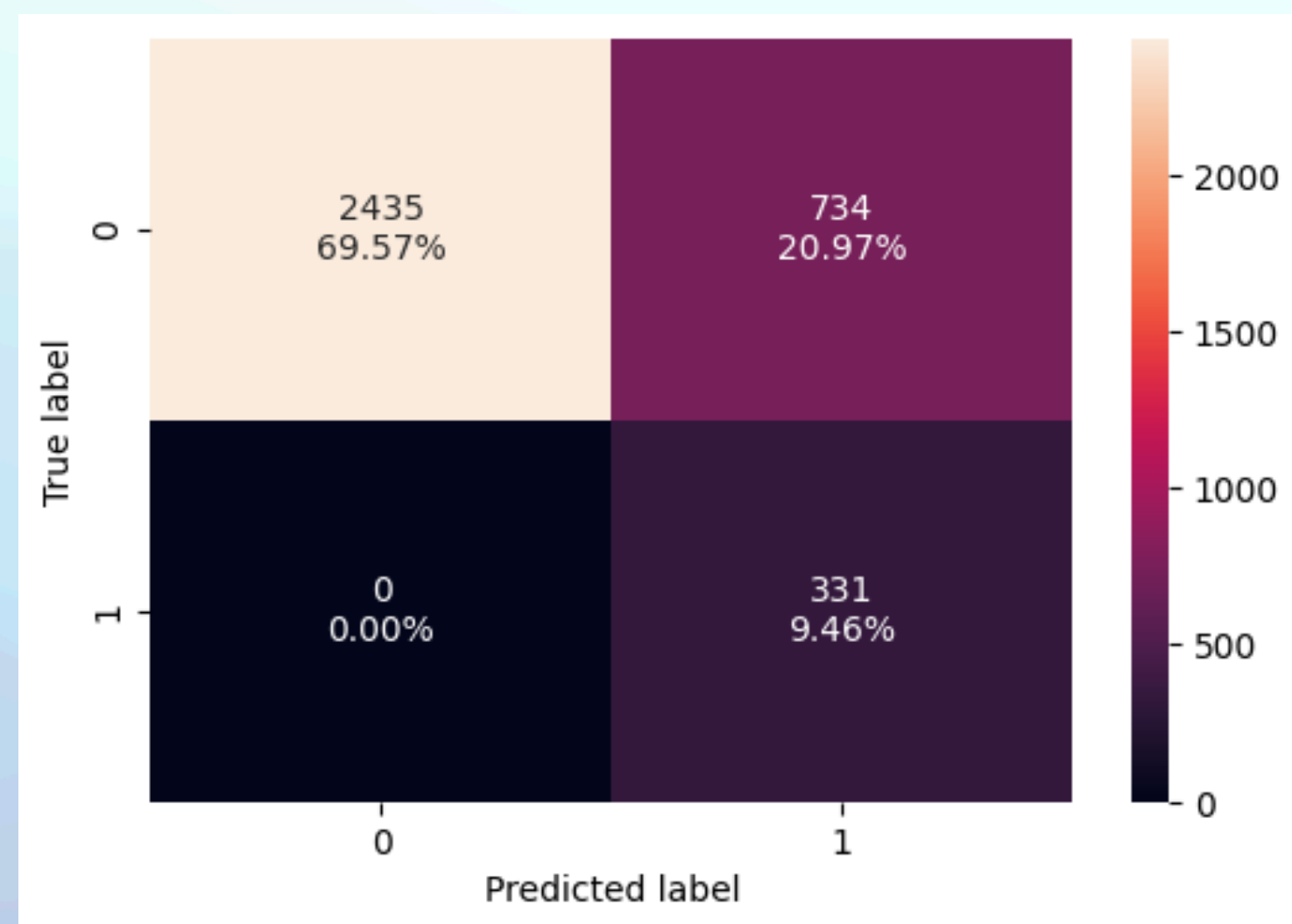


Model Performance Improvement

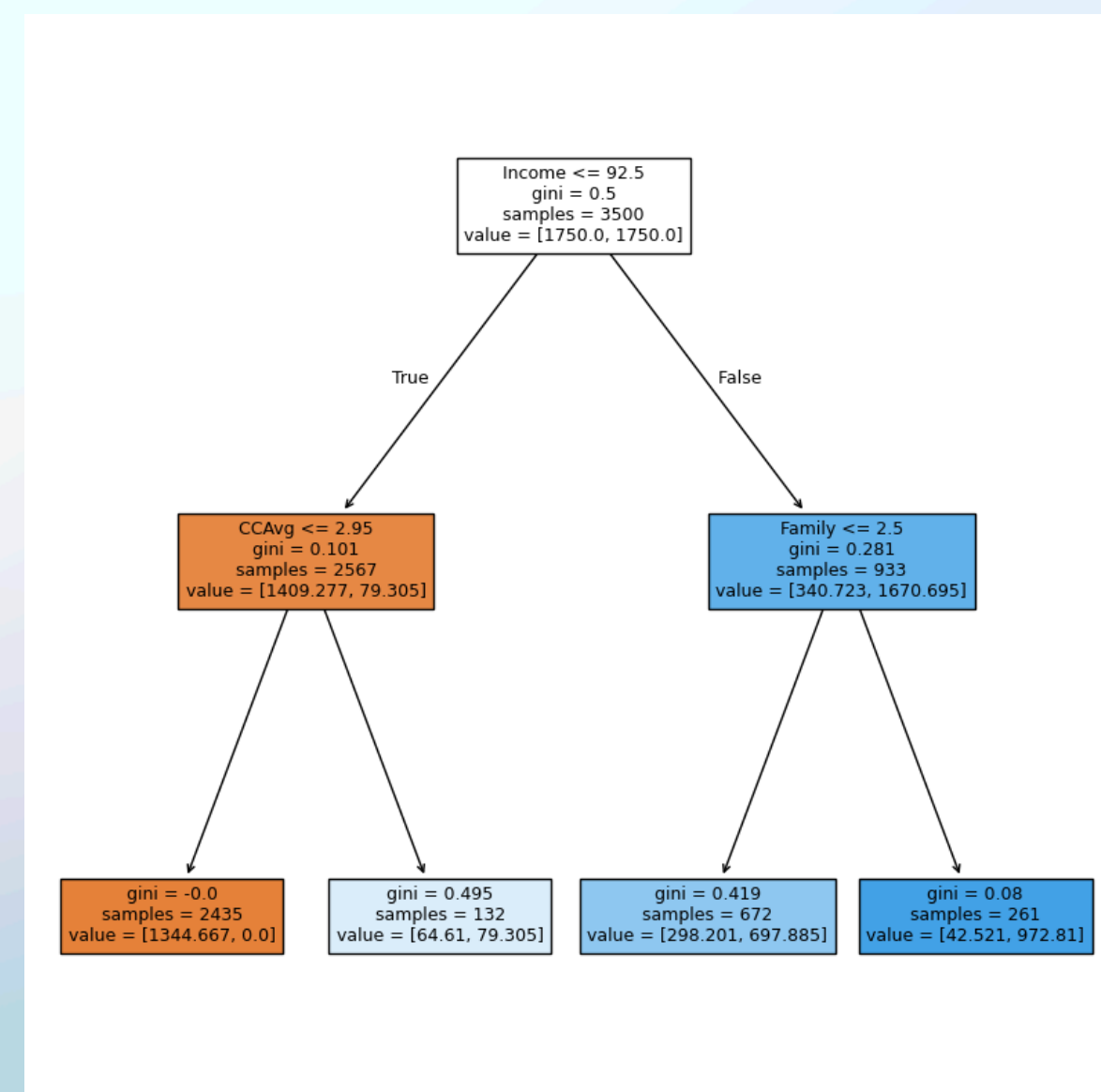
Pre-Pruning - Training Data & Decision Tree

We see after Pre-pruning we moved the model away from 'memorizing' data to a more realistic and balanced model. We are still capturing all potential borrowers and our recall is at 100%.

We do however see there are 734 false positives and we are over targeting people who will likely not want a loan.



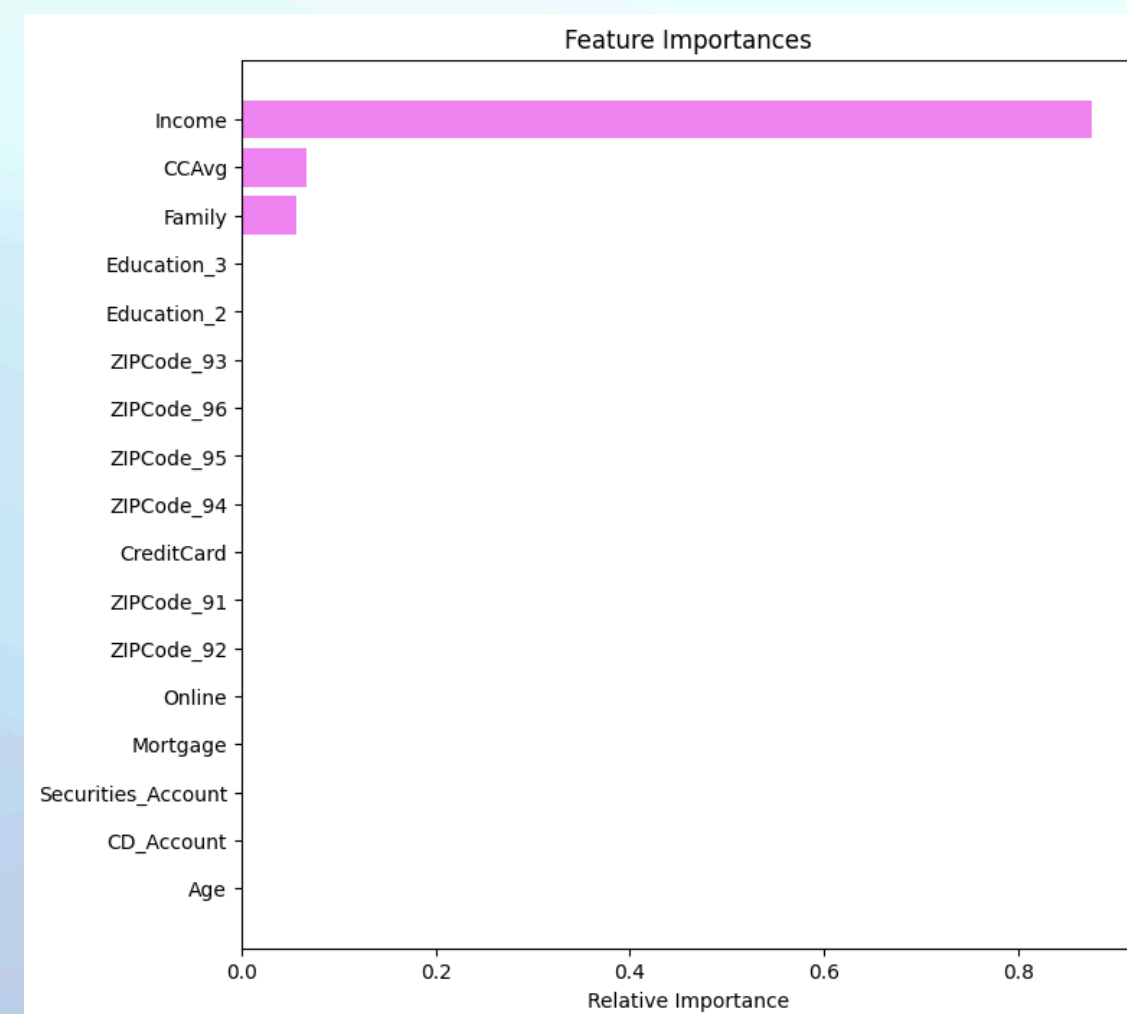
We see that our decision tree is simple and has two levels of depth. The blue boxes represent the groups that are identified as likely to borrow.



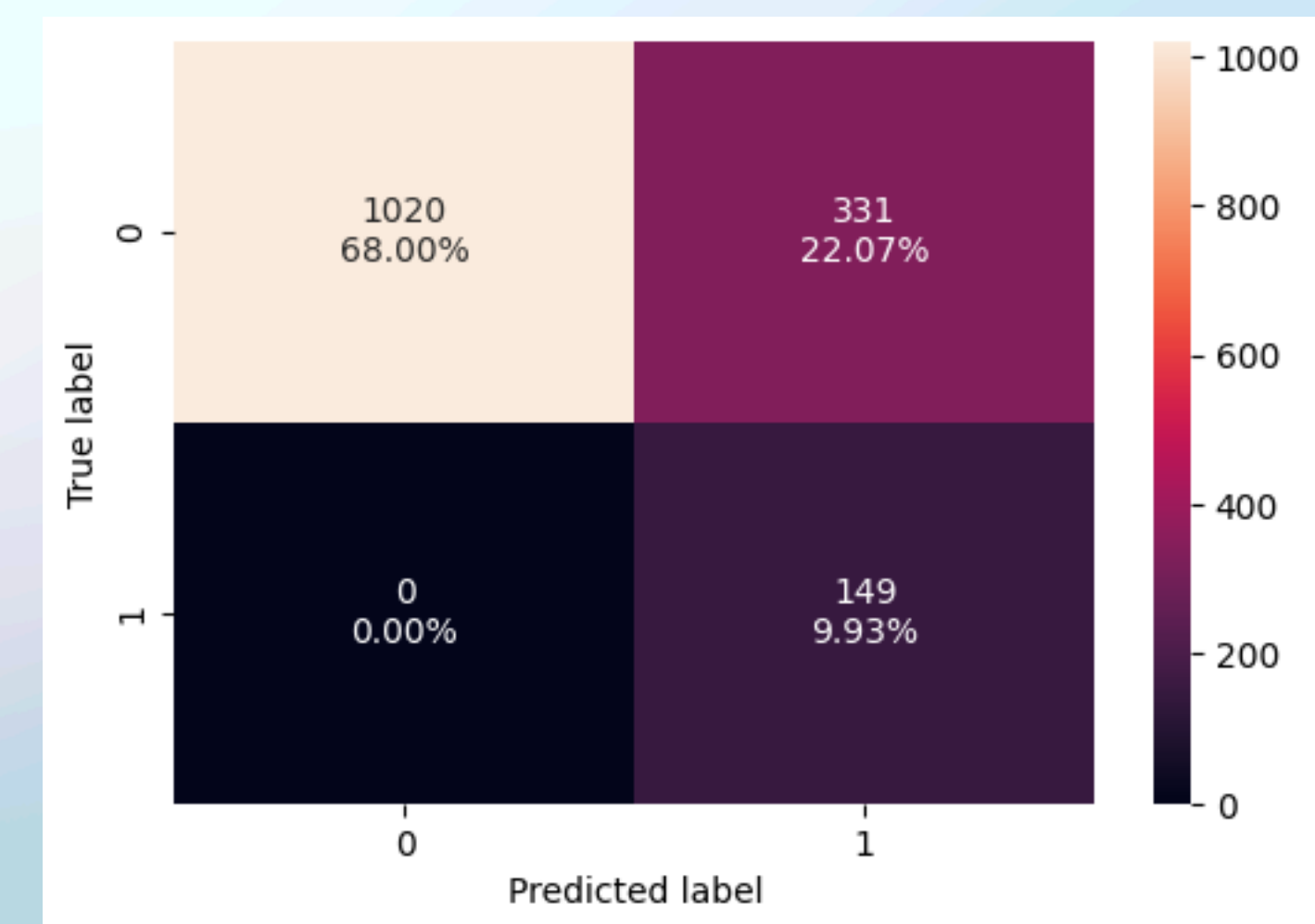
Model Performance Improvement

Pre-Pruning - Feature Importances & Testing Data

Next we will look at the feature importances. We see that Income is the most important feature followed by CCAvg and Family.



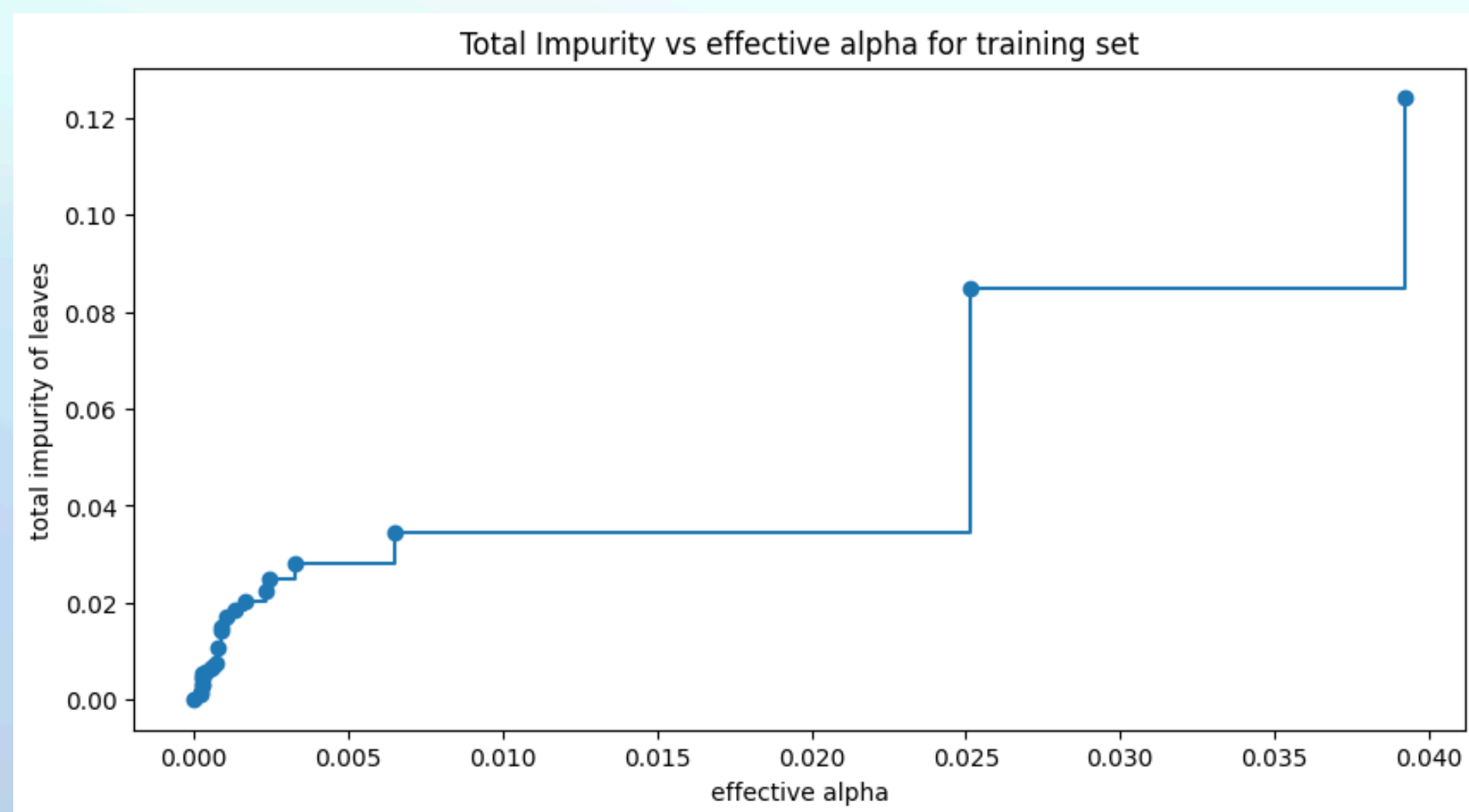
We see that our Testing Data is perfect with its recall. There are no false negatives in this Training data which means the model is not missing a single potential borrower.



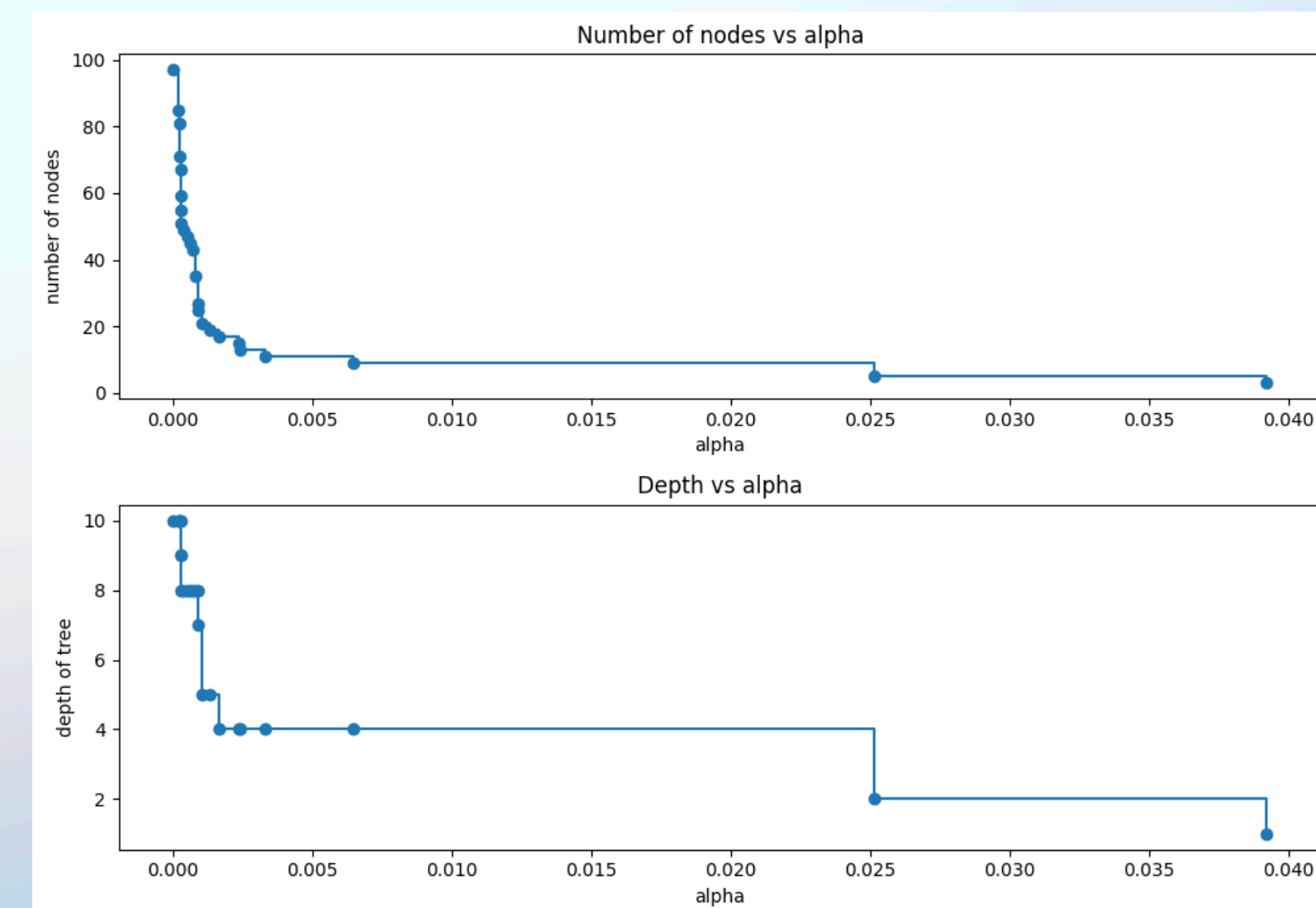
Model Performance Improvement

Post-Pruning - Effective Alphas

Next we will look at the Total Impurity vs effective alpha for the training set. We see that the optimal zone is from 0.005 - 0.010 because it is relatively flat which means an alpha in this range removes the 'noise' in the data but keeps important patterns like 'Income' and 'Family Size'.



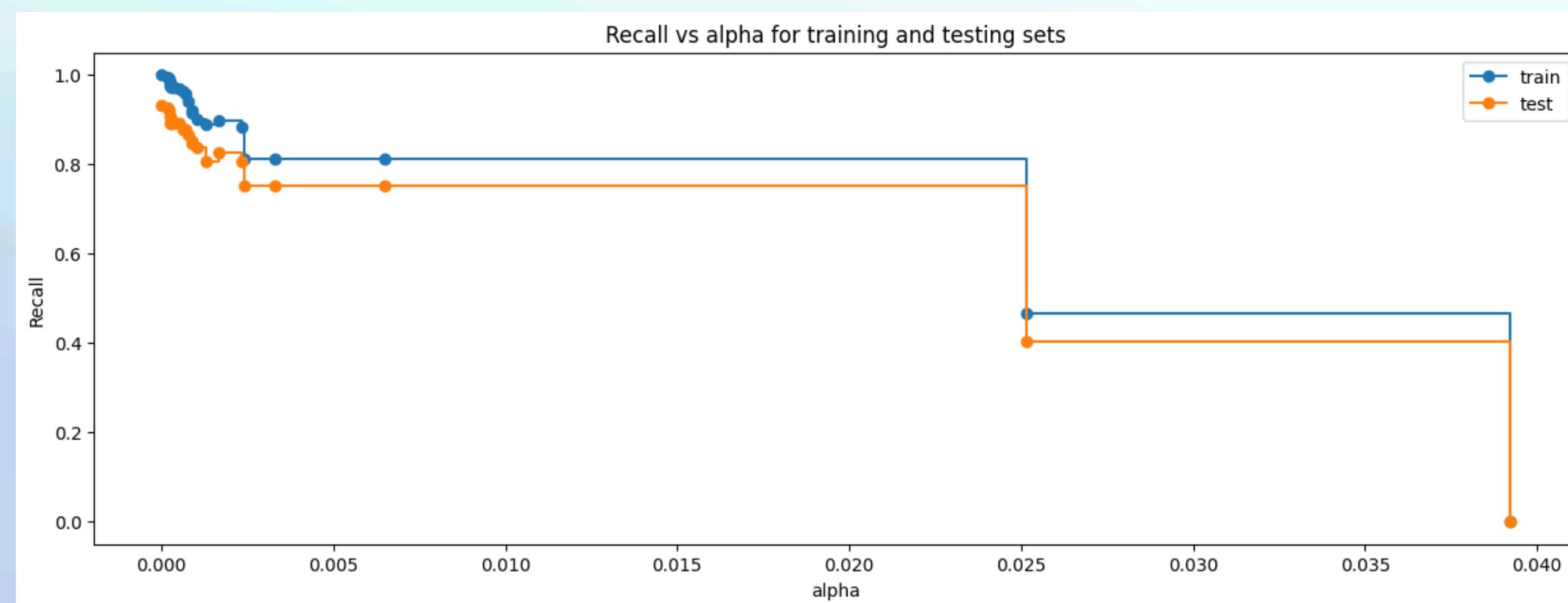
We see that in the top plot the part that plateau's is between 0.005 - 0.025 and this suggests that the model has found a stable core of rules that is ideal for our model. On the bottom plot we see from 0.003-0.025 there is a plateau which means the model is very reliable.



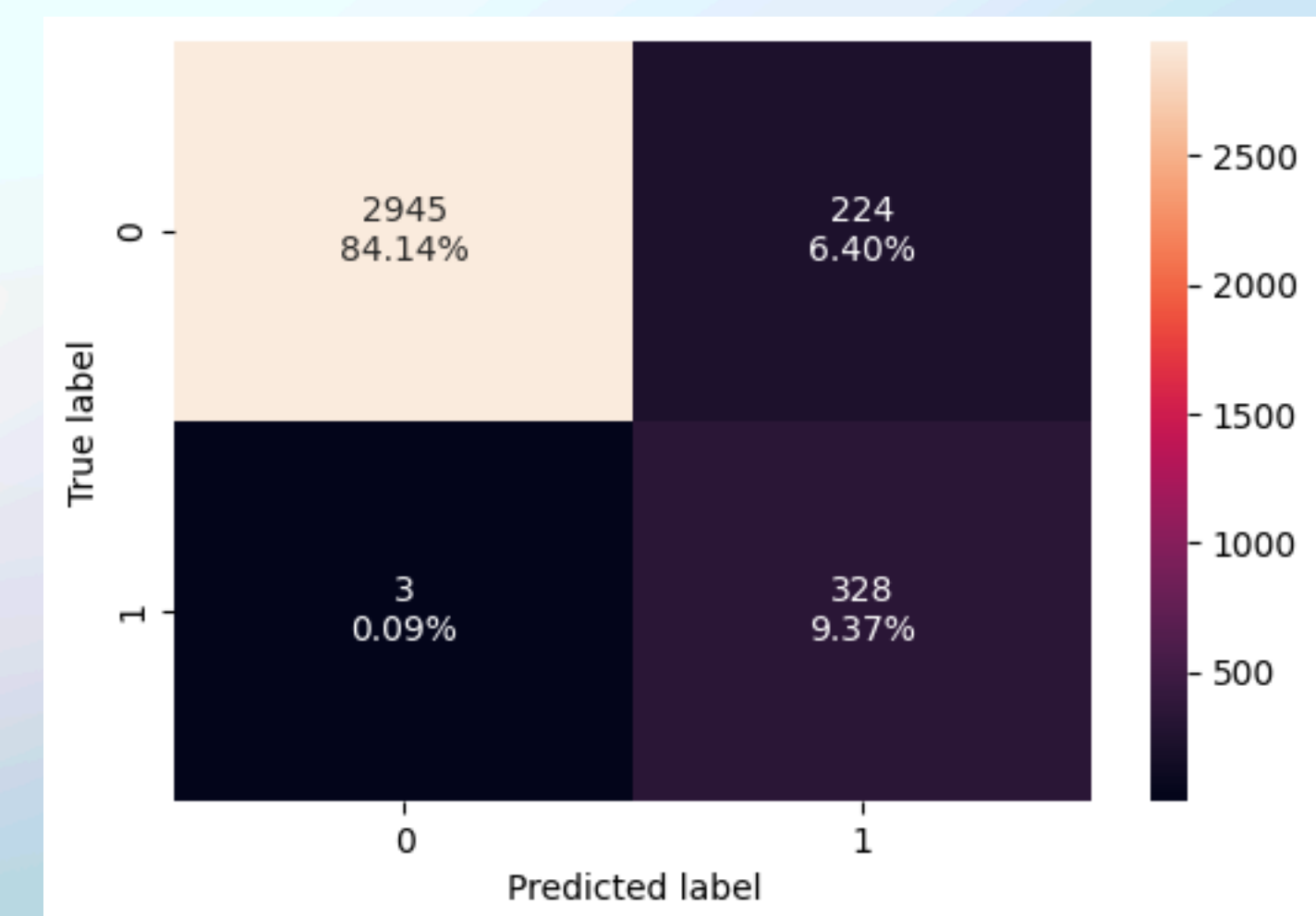
Model Performance Improvement

Post-Pruning

On the chart below we are comparing the recall vs alpha for the training and testing sets. We see that the training recall starts at 1.0 and the test recall starts at .90. This means our model is 'memorizing' specific people in the training set who do not exist in the test set we should pick an alpha around 0.005 - 0.015 for our model because this model is stable and will give us the best results.



We see that our model has excellent recall and is finding almost every potential borrower but is missing a few. It has a very high accuracy which means the model is very reliable. The 224 false positives are good people to go after because they look like they should be borrowers but they said no to a personal loan.

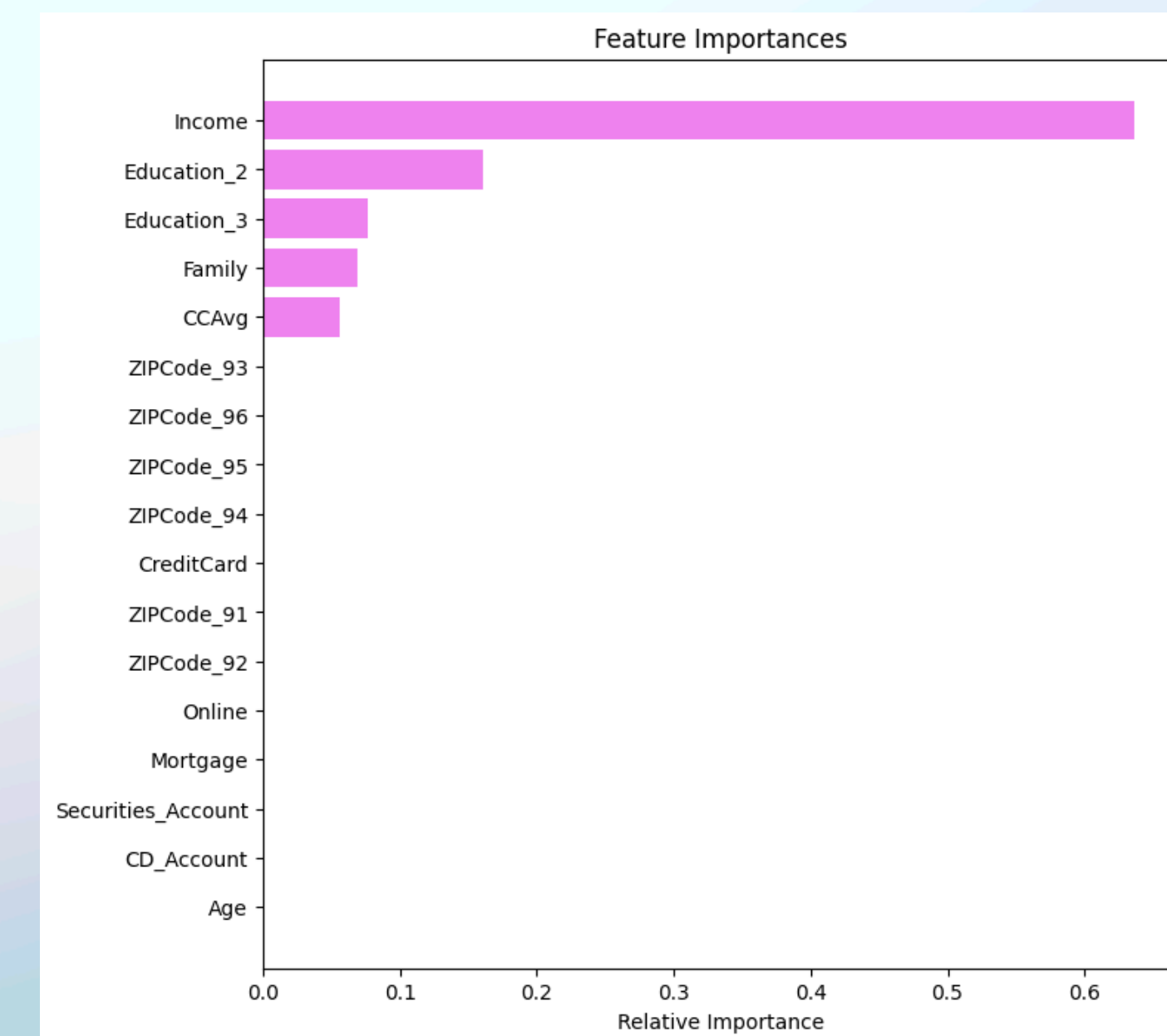
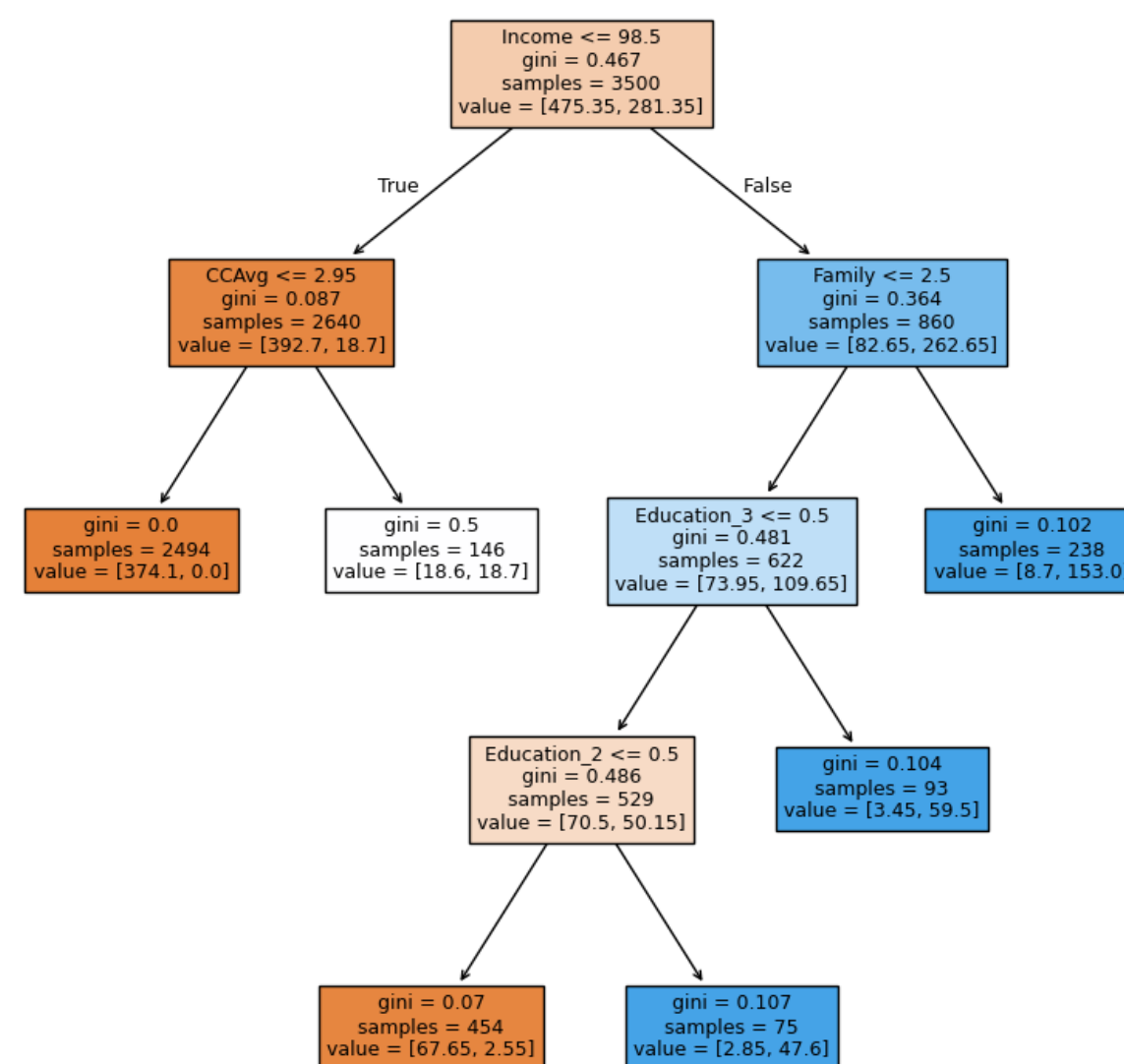


Model Performance Improvement

Post-Pruning

We see our decision tree after post pruning it. We can see that the root node $\text{Income} \leq 98.5$ is the most important factor. Our model is telling us that people with $\text{Income} > 98.5$ and family size > 2.5 are highly likely to take a loan. It is also saying that people with $\text{Income} > 98.5$ and education level 3 or likely to take out loans too.

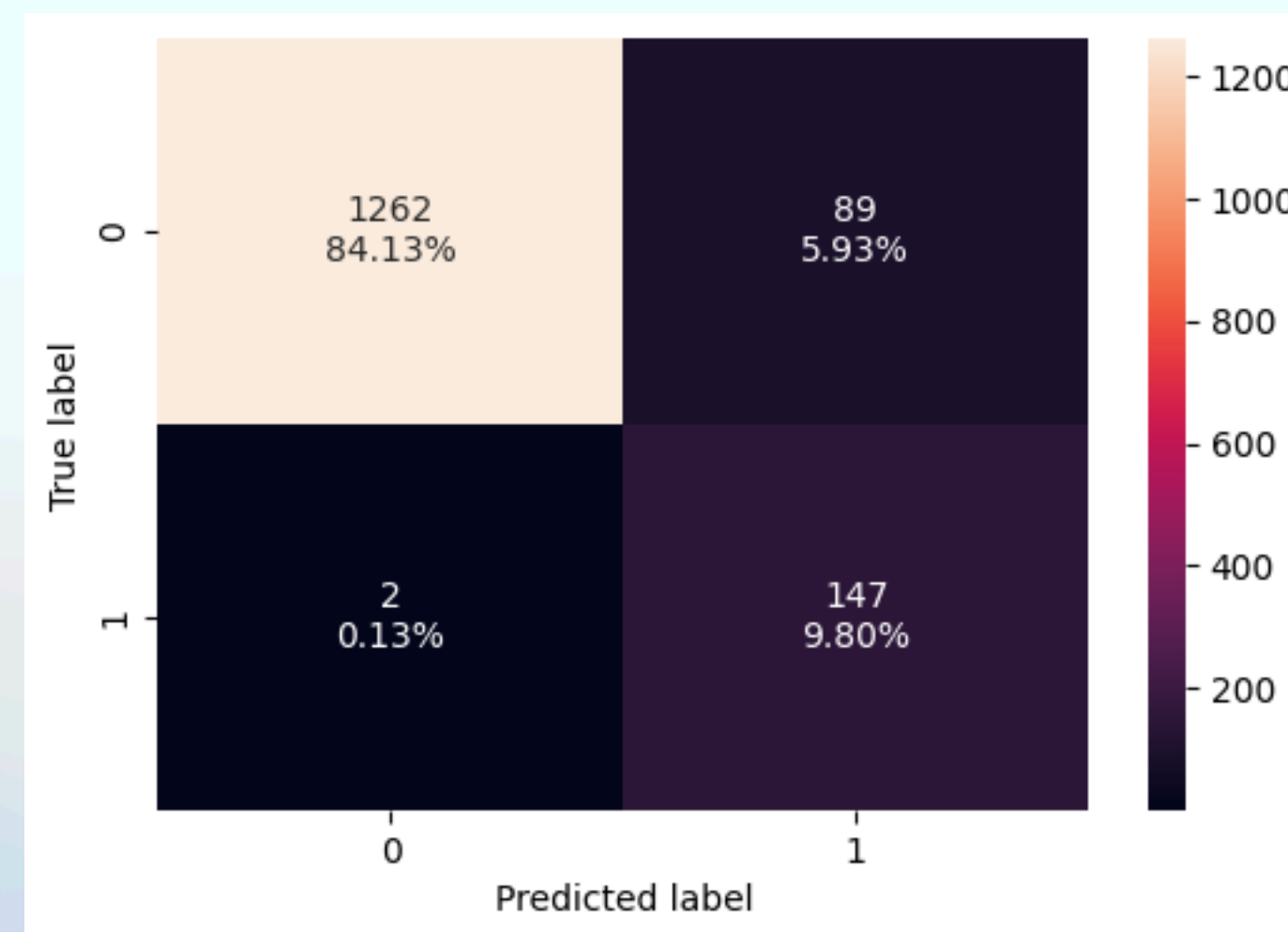
We see that income, Education 2&3, Family and CCAvg are the most important features of this model.



Model Performance Improvement

Post-Pruning - Checking Performance on Test Data

We see We see the recall on our model is very good. We only missed 2 borrowers. The model does predict 89 people to take out a loan which they said no. These are people we can target for the future since they fit the high income and large family criteria. Our model correctly identified 1,262 people who said no to a loan and this saves the company time because they do not need to be contacted about a loan.



Model Performance Improvement

Pre-Pruning vs Post-Pruning - Training and Testing Data

In the training data performance comparison we see that the pre-pruning decision tree Accuracy is 0.79, Recall is 1.00, Precision is 0.31 and F1 is 0.47.

In the post-pruned decision tree we see Accuracy is 0.93, Recall is 0.99, Precision is 0.59 and F1 is 0.74.

The default decision tree is at 1.0 for Accuracy, Recall, Precision and F1.

In the testing data performance comparison we see that the pre-pruning decision tree Accuracy is 0.77, Recall is 1.00, Precision is 0.31 and F1 is 0.47.

In the post-pruned decision tree we see Accuracy is 0.93, Recall is 0.98, Precision is 0.62 and F1 is 0.76.

The default decision tree is at Accuracy 0.98, Recall 0.93, Precision 0.92 and F1 0.92

Which Model is Best?

The post-pruning model is the best model to choose. The reasons why are:

Pre-Pruning vs Post-Pruning

- Because the post-pruned model only had 89 false positives compares to 734 false positives in the pre-pruned model
- We reduced calls by over 80% only targeting the best candidates rather than calling 1,000 people and majority of them are not good candidates
- Post-pruning lets you see the full picture and then you can remove the branches that are not needed. The model still maintained a Recall of 98-99% after post-pruning
- Overall the post-pruning model saves AllLife Bank time by reducing the calls needed to be made and identifies the best candidates for a personal loan